MONOTONICITY OF ERDŐS SUMS

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ABSTRACT. For $k \geq 1$ let $f_k = \sum_{\Omega(n)=k} 1/(n \log n)$, where Ω counts prime factors with multiplicity. We prove that $f_k < f_{k+1}$ for every $k \geq 6$, whereas the sums g_k of $1/(n \log n)$ over odd n with $\Omega(n) = k$ satisfy $g_k > g_{k+1}$ for every $k \geq 1$. The two statements arise from the same analytic distinction. Writing the generating function $\sum_k P_k(s) z^k$ of the associated Dirichlet series as $\zeta(s)^z H(s, z)$, at $s = 1$ the point $z = 2$ lies on the first singular circle for the unrestricted product, and strictly inside the first singular circle of the product with the prime 2 removed. The analytic reductions are written explicitly below. The finite inequalities and coefficient bounds used in the last step are certified by ball arithmetic, and the reductions themselves are machine-checked in Lean 4 on top of those certified inputs; the separation between analytic argument and certified input is stated precisely, and tabulated.

1. INTRODUCTION

For $k \geq 1$ put

$$f_k = \sum_{\Omega(n)=k} \frac{1}{n \log n}.$$

Erdős [4] proved that the corresponding sum is bounded uniformly over primitive sets. In particular each f_k converges. Zhang [7] proved $f_k < f_1$ for $k > 1$, and Bayless, Kinlaw and Klyve [8] proved $f_2 > f_3$. Banks and Martin [1] conjectured that

$$f_1 > f_2 > f_3 > \cdots. \tag{1}$$

Lichtman [2] disproved (1), showing that $f_k \to 1$ and that the sequence attains a global minimum at $k = 6$. Gorodetsky, Lichtman and Wong [3] then established

$$f_k = 1 - 2^{-k}\big(a_1 k^2 + O(k \log(k+1))\big), \qquad a_1 = \frac{d \log 2}{4} > 0, \tag{2}$$

where

$$d = \frac{1}{4} \prod_{p \geq 3} \left(1 - \frac{2}{p}\right)^{-1} \left(1 - \frac{1}{p}\right)^{2}. \tag{3}$$

Neither of these yields monotonicity beyond the minimum. A global minimum at $k = 6$ leaves open whether $f_k < f_{k+1}$ holds at every index above it, and the implicit constant in (2) does not make explicit a range on which the increase can be read off. Our first theorem gives that monotonicity in full.

Theorem 1.1. *For every $k \geq 6$ one has $f_k < f_{k+1}$.*

Let

$$g_k = \sum_{\substack{n \text{ odd} \\ \Omega(n)=k}} \frac{1}{n \log n}.$$

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The second theorem shows that deleting the prime 2 reverses the direction of monotonicity at every index.

Theorem 1.2. *For every $k \geq 1$ one has $g_k > g_{k+1}$, and $g_k \to \frac{1}{2}$.*

The proof of Theorem 1.2 gives the more precise expansion

$$g_k = \frac{1}{2} + \beta 2^{-k} + R_k, \qquad |R_k| \leq 4900(k+3)^3 3^{-k} \quad (k \geq 59), \tag{4}$$

where $\gamma = 0.5772156649\ldots$ is Euler's constant and

$$\beta = d\left(\gamma + \log 2 - \sum_{p \geq 3} \frac{\log p}{(p-1)(p-2)}\right) > 0. \tag{5}$$

The 2^{-k} term comes from evaluating a regularized Euler product at $z = 2$. For odd integers the first pole is at $z = 3$, so this evaluation takes place strictly inside the disc of analyticity. For the unrestricted sums the first pole is at $z = 2$ itself. The boundary contribution then acquires the factor k^2 in (2).

The proof has two kinds of input. Identities, coefficient reductions and inequalities are proved in the text. Bounds produced by interval arithmetic are isolated as certified propositions. Thus a sentence of the form "the computation gives" never substitutes for an analytic step: the exact inequality supplied by the computation is stated before it is used.

2. Generating functions and the analytic boundary

For $s > 1$ define

$$P_k(s) = \sum_{\Omega(n)=k} n^{-s}, \qquad P_k^{\mathrm{odd}}(s) = \sum_{\substack{n \text{ odd} \\ \Omega(n)=k}} n^{-s}.$$

Lemma 2.1. *For every $k \geq 1$,*

$$f_k = \int_1^\infty P_k(s)\,\mathrm{d}s, \qquad g_k = \int_1^\infty P_k^{\mathrm{odd}}(s)\,\mathrm{d}s. \tag{6}$$

Proof. For $n \geq 2$,

$$\int_1^\infty n^{-s}\,\mathrm{d}s = \int_1^\infty \mathrm{e}^{-s\log n}\,\mathrm{d}s = \left[-\frac{\mathrm{e}^{-s\log n}}{\log n}\right]_{s=1}^{\infty} = \frac{1}{n\log n}.$$

All summands are non-negative. Tonelli's theorem therefore permits the sum and integral to be interchanged in both cases, giving

$$\int_1^\infty \sum_{\Omega(n)=k} n^{-s}\,\mathrm{d}s = \sum_{\Omega(n)=k} \int_1^\infty n^{-s}\,\mathrm{d}s = f_k,$$

and the same calculation with the sum restricted to odd n proves the second identity. □

The generating functions are

$$\sum_{k\geq 0} P_k(s) z^k = \prod_p (1 - zp^{-s})^{-1}, \qquad \sum_{k\geq 0} P_k^{\mathrm{odd}}(s) z^k = \prod_{p\geq 3} (1 - zp^{-s})^{-1}. \tag{7}$$

Write these products as $\zeta(s)^z H(s,z)$ and $\zeta(s)^z H_{\mathrm{o}}(s,z)$, where

$$H(s,z) = \prod_p (1 - zp^{-s})^{-1}(1-p^{-s})^z, \tag{8}$$

$$H_{\mathrm{o}}(s,z) = (1-2^{-s})^z \prod_{p\geq 3} (1 - zp^{-s})^{-1}(1-p^{-s})^z. \tag{9}$$

Proposition 2.2. *Let $\sigma = \operatorname{Re} s$. Then H is analytic in the two complex variables (s,z) on $\{\sigma > 1/2,\ |z| < 2^\sigma\}$, and H_{o} on $\{\sigma > 1/2,\ |z| < 3^\sigma\}$. In particular, for fixed real $s > 1/2$ the function $H(s,\cdot)$ is analytic in $|z| < 2^s$ and $H_{\mathrm{o}}(s,\cdot)$ in $|z| < 3^s$; each $\partial_s^r H_{\mathrm{o}}(s,\cdot)$ is again analytic in $|z| < 3^s$, and*

$$\frac{\mathrm{d}^r}{\mathrm{d}s^r}[z^j]H_{\mathrm{o}}(s,z) = [z^j]\partial_s^r H_{\mathrm{o}}(s,z); \tag{10}$$

the same holds for H in $|z| < 2^s$.

Proof. Let $x = p^{-s}$, so that $|x| = p^{-\sigma} \leq 2^{-\sigma} < 2^{-1/2}$ on the domain. The factor $(1-x)^z$ is a non-integer power of a complex number, so a branch must be fixed; the principal one will do throughout, since $1-x$ then lies in the disc of radius $2^{-1/2} < 1$ about 1, on which the principal logarithm is analytic and where no branch cut is met. With that convention, for (s,z) in a compact subset of the domain — so that $\sigma \geq s_0 > 1/2$ and $|z| \leq R < 2^{s_0}$, whence $|zx| < 1$ for every p — the logarithm of the pth regularized factor is

$$\begin{aligned} -\log(1-zx) + z\log(1-x) &= \sum_{m\geq 1} \frac{z^m - z}{m} x^m \\ &= \sum_{m\geq 2} \frac{z^m - z}{m} x^m = O_K(p^{-2\sigma}), \end{aligned}$$

the $m = 1$ terms cancelling. The estimate is uniform for $z \in K$ and depends on s only through σ. Since $\sum_p p^{-2\sigma} < \infty$ for $\sigma > 1/2$, the product of the regularized factors converges locally uniformly wherever none of the factors has a pole. The poles occur at $z = p^s$, of modulus p^σ; the nearest has modulus 2^σ for (8) and 3^σ for (9). A locally uniform limit of analytic functions is analytic, which proves the first assertion.

Differentiability in s does not follow from this if s is held real: uniform convergence of real-variable functions says nothing about the derivative of the limit. What is used instead is that the variable s may be taken complex. Each $p^{-s} = \mathrm{e}^{-s\log p}$ is entire in s, so every regularized factor is analytic in (s,z) jointly wherever its denominator does not vanish, which on $\{|z| < 3^\sigma\}$ it does not, since $|z| < 3^\sigma \leq p^\sigma = |p^s|$ for every $p \geq 3$. The estimate above depends on s only through $|p^{-s}| = p^{-\sigma}$, so it holds uniformly on any set with $\sigma \geq s_0 > 1/2$, and $\sum_p p^{-2s_0} < \infty$. The product therefore converges locally uniformly on the stated domain in $\mathbb{C}^2$, and a locally uniform limit of analytic functions of several complex variables is analytic; this is the first assertion. Analyticity in s gives derivatives of every order at real s, and since $[z^j]$ is a Cauchy integral over a circle that may be held fixed on a compact subset, the two operations commute, which is (10). □

At $s = 1$ the point $z = 2$ lies on the first singular circle for H and lies strictly inside the first singular circle for H_{o}. This is the analytic distinction used below.

We shall also need the local change of variables used in the odd case.

Lemma 2.3. *Let $u = \log\zeta(s)$ for $s > 1$, and let $s = s(u)$ be the inverse function. Put $\varepsilon(u) = s(u) - 1$ and $J(u) = -s'(u)$. As $u \to \infty$,*

$$\varepsilon(u) = \mathrm{e}^{-u} + \gamma\mathrm{e}^{-2u} + O(\mathrm{e}^{-3u}), \qquad J(u) = \mathrm{e}^{-u} + 2\gamma\mathrm{e}^{-2u} + O(\mathrm{e}^{-3u}). \tag{11}$$

Proof. The point of the argument is that the expansions below are convergent power series and not merely asymptotic, which is what allows the second of them to be differentiated term by term.

Since ζ is meromorphic on $\mathbb{C}$ with a single simple pole, of residue 1, at $s = 1$, the function $\varepsilon \mapsto \varepsilon\,\zeta(1+\varepsilon)$ is analytic near $\varepsilon = 0$ with value 1 there; its expansion is $1 + \gamma\varepsilon + O(\varepsilon^2)$. Hence

$$x = \frac{1}{\zeta(1+\varepsilon)} = \frac{\varepsilon}{\varepsilon\,\zeta(1+\varepsilon)}$$

is analytic near $\varepsilon = 0$, with $x(0) = 0$ and $x = \varepsilon - \gamma\varepsilon^2 + O(\varepsilon^3)$, so in particular $\mathrm{d}x/\,\mathrm{d}\varepsilon = 1 \neq 0$ at the origin. By the inverse function theorem for analytic functions, ε is therefore an analytic function of x near $x = 0$, given by a convergent power series

$$\varepsilon = \psi(x) = x + \gamma x^2 + O(x^3),$$

which is the first formula in (11) once $x = \mathrm{e}^{-u}$. Because ψ is analytic rather than merely asymptotic, it may be differentiated termwise: $\psi'(x) = 1 + 2\gamma x + O(x^2)$. With $x = \mathrm{e}^{-u}$ one has $\mathrm{d}x/\,\mathrm{d}u = -x$, so the chain rule gives

$$\varepsilon'(u) = \psi'(x)\frac{\mathrm{d}x}{\mathrm{d}u} = -x\big(1 + 2\gamma x + O(x^2)\big) = -x - 2\gamma x^2 + O(x^3),$$

and $J(u) = -\varepsilon'(u) = x + 2\gamma x^2 + O(x^3)$. □

3. The odd sums

Throughout this section write

$$h_j(s) = [z^j]H_{\mathrm{o}}(s,z), \qquad b_j = h_j'(1) + 2\gamma h_j(1).$$

Since $\zeta(s)^z = \sum_{m\geq 0}(\log\zeta(s))^m z^m/m!$, coefficient extraction in (7) gives the exact identity

$$P_k^{\mathrm{odd}}(s) = \sum_{j=0}^{k} h_j(s)\frac{(\log\zeta(s))^{k-j}}{(k-j)!}. \tag{12}$$

Using $u = \log\zeta(s)$ in Lemma 2.1, and noting that s decreases from ∞ to 1 as u increases from 0 to ∞, we obtain

$$g_k = \sum_{j=0}^{k}\int_0^\infty \frac{u^{k-j}}{(k-j)!}\,h_j(s(u))J(u)\,\mathrm{d}u. \tag{13}$$

3.1. The first two terms.

Proposition 3.1. *With h_j and b_j as above and E_j as in the proof below, put*

$$R_k = \sum_{j=0}^{k}\int_0^\infty \frac{u^{k-j}}{(k-j)!}E_j(u)\,\mathrm{d}u - \sum_{j>k} h_j(1) - 2^{-k-1}\sum_{j>k} b_j 2^j. \tag{14}$$

Then

$$g_k = H_{\mathrm{o}}(1,1) + 2^{-k-1}\big(\partial_s H_{\mathrm{o}}(1,2) + 2\gamma H_{\mathrm{o}}(1,2)\big) + R_k. \tag{15}$$

Proof. Let $x = \mathrm{e}^{-u}$. By Lemma 2.3, $J(u) = x + 2\gamma x^2 + O(x^3)$ and $\varepsilon(u)J(u) = x^2 + O(x^3)$. Taylor's theorem at $s = 1$ gives

$$h_j(1+\varepsilon) = h_j(1) + \varepsilon h_j'(1) + \frac{\varepsilon^2}{2}h_j''(1+\theta_{j,u}\varepsilon)$$

for some $0 < \theta_{j,u} < 1$. Put

$$r_0(u) = J(u) - \mathrm{e}^{-u} - 2\gamma\mathrm{e}^{-2u}, \qquad r_1(u) = \varepsilon(u)J(u) - \mathrm{e}^{-2u}.$$

Multiplication by $J(u)$ now gives the exact identity

$$\begin{aligned} h_j(s(u))J(u) &= h_j(1)\mathrm{e}^{-u} + b_j\mathrm{e}^{-2u} + E_j(u), \\ E_j(u) &= h_j(1)r_0(u) + h_j'(1)r_1(u) + \frac{\varepsilon(u)^2 J(u)}{2} h_j''(1 + \theta_{j,u}\varepsilon(u)). \end{aligned} \tag{16}$$

Thus no error is suppressed in the notation E_j: its three summands are the Jacobian error, the first-order coordinate error and the Taylor remainder.

Insert (16) into (13). Since

$$\int_0^\infty \frac{u^m}{m!}\mathrm{e}^{-ru}\,\mathrm{d}u = r^{-m-1} \qquad (r > 0),$$

the first two terms become

$$\sum_{j=0}^{k} h_j(1) = H_{\mathrm{o}}(1,1) - \sum_{j>k} h_j(1), \tag{17}$$

$$\sum_{j=0}^{k} b_j 2^{-(k-j+1)} = 2^{-k-1}\big(\partial_s H_{\mathrm{o}}(1,2) + 2\gamma H_{\mathrm{o}}(1,2)\big) - 2^{-k-1}\sum_{j>k} b_j 2^j. \tag{18}$$

Substituting (17) and (18), and collecting the E_j integrals with the two tails, is exactly (14); this gives (15). Note that (14) is an identity, not an error term absorbed into an O: every later estimate is an estimate of those three explicit quantities. □

Lemma 3.2. *One has*

$$H_{\mathrm{o}}(1,1) = \frac{1}{2}, \qquad H_{\mathrm{o}}(1,2) = d, \tag{19}$$

and

$$\partial_s \log H_{\mathrm{o}}(1,2) = 2\log 2 - 2\sum_{p\geq 3} \frac{\log p}{(p-1)(p-2)}. \tag{20}$$

Proof. At $s = z = 1$, every factor in the product over $p \geq 3$ cancels, and hence

$$H_{\mathrm{o}}(1,1) = (1 - 2^{-1}) = \frac{1}{2}.$$

At $(s,z) = (1,2)$,

$$H_{\mathrm{o}}(1,2) = \frac{1}{4}\prod_{p\geq 3}\left(1 - \frac{2}{p}\right)^{-1}\left(1 - \frac{1}{p}\right)^2 = d.$$

For the derivative, take logarithms:

$$\log H_{\mathrm{o}}(s,2) = 2\log(1 - 2^{-s}) + \sum_{p\geq 3}\left[-\log(1 - 2p^{-s}) + 2\log(1 - p^{-s})\right].$$

The differentiated pth summand is $O((\log p)p^{-2s})$ locally uniformly near $s = 1$, so termwise differentiation is permitted. At $s = 1$ the first term contributes $2\log 2$, while the pth summand contributes

$$-\frac{2\log p}{p-2} + \frac{2\log p}{p-1} = -\frac{2\log p}{(p-1)(p-2)}.$$

Summing over $p \geq 3$ proves (20). □

Combining Proposition 3.1 and Lemma 3.2 gives

$$g_k = \frac{1}{2} + \beta 2^{-k} + R_k, \qquad \beta = d\left(\gamma + \log 2 - \sum_{p\geq 3} \frac{\log p}{(p-1)(p-2)}\right). \tag{21}$$

3.2. **The remainder.** Three weighted ℓ^1 norms of coefficients carry all of the interval input for the odd sums. Put

$$\begin{aligned} \Sigma_0 &= \sum_{m\geq 0} 3^m |[z^m](1-z/3)H_{\mathrm{o}}(1,z)|, \\ \Sigma_1 &= \sum_{m\geq 0} 3^m |[z^m](1-z/3)^2 \partial_s H_{\mathrm{o}}(1,z)|, \\ \Sigma_2 &= \sup_{1\leq s\leq s(3)} \sum_{m\geq 0} 3^m |[z^m](1-z/3)^3 \partial_s^2 H_{\mathrm{o}}(s,z)|. \end{aligned} \tag{22}$$

Finiteness of Σ_0 and Σ_1 is analytic, and the weights $r = 1, 2$ on the factors $(1-z/3)^r$ are what makes it so. Proposition 2.2 gives $H_{\mathrm{o}}(1,\cdot)$ only on $|z| < 3$, which by itself would not make a sum weighted by 3^m converge. But at $s = 1$ the nearest pole in z is at $z = 3$, contributed by the single Euler factor $(1-z/3)^{-1}$, and after $r-1$ derivatives in s that pole has order r; multiplying by $(1-z/3)^r$ cancels it exactly, so both functions continue analytically to $|z| < 5$, the next pole being at $z = 5$. Their coefficients decay like 5^{-m} and the weighted sums converge geometrically.

For Σ_2 the same device applies only at the left endpoint. At $s > 1$ the pole sits at $z = 3^s$, which $(1-z/3)^3$ does not cancel, and the inner sum is finite because the radius of convergence is then $3^s > 3$ — a margin that shrinks to nothing as $s \to 1^+$, so this does not by itself bound the supremum. Each inner sum is finite for every s in the range, by cancellation at $s = 1$ and by $3/3^s < 1$ for $s > 1$; the finiteness of the supremum over $[1, s(3)]$ is part of what the interval computation below establishes, not a separate analytic claim. For Σ_0 and Σ_1 only the numerical values are certified.

Next, split the remainder of Proposition 3.1 at $u = 3$: with E_j the explicit expression in (16), put

$$R_{k,\mathrm{far}} = \sum_{j=0}^{k} \int_3^\infty \frac{u^{k-j}}{(k-j)!} E_j(u)\,\mathrm{d}u, \qquad R_{k,\mathrm{near}} = \sum_{j=0}^{k} \int_0^3 \frac{u^{k-j}}{(k-j)!} E_j(u)\,\mathrm{d}u, \tag{23}$$

so that R_k is $R_{k,\mathrm{far}} + R_{k,\mathrm{near}}$ together with the two coefficient tails of (17)–(18). Here $s(3)$ denotes the value of s at $u = 3$.

The far range needs one further constant. Write $\varepsilon = s - 1$ and $J = \zeta(s)/(-\zeta'(s))$; this is the J of Lemma 2.3 expressed in s rather than u, since $\mathrm{d}u/\,\mathrm{d}s = \zeta'(s)/\zeta(s)$ makes $-s'(u)$ equal to $\zeta(s)/(-\zeta'(s))$, and the four quantities below compare it with powers of $\zeta^{-1} = \mathrm{e}^{-u}$. Let

$$\begin{gathered} C_b = \sup|\varepsilon\zeta|, \qquad C_f = \sup|\zeta^2\big(J - \zeta^{-1}\big)|, \\ C_J = \sup\big|\zeta^3\big(J - \zeta^{-1} - 2\gamma\zeta^{-2}\big)\big|, \qquad C_P = \sup\big|\zeta^3\big(\varepsilon J - \zeta^{-2}\big)\big|, \\ C_E = C_J + C_P + C_b^2\,(C_f + 1), \end{gathered} \tag{24}$$

every supremum being over $1 < s \leq s(3)$. Each of the four is a ratio in which the pole of ζ at $s = 1$ cancels. How they are computed matters, and is described in Section 8: evaluating them on boxes of s loses that cancellation and the enclosures diverge however finely the range is partitioned, so a Taylor model is used instead.

Proposition 3.3 (Certified odd input)**.** *With Σ_i as in* (22) *and C_E as in* (24)*, the following are certified by interval arithmetic.*
(i) The coefficient norms satisfy

$$\Sigma_0 \le 44.5987, \qquad \Sigma_1 \le 486.1946, \qquad \Sigma_2 \le 14414, \qquad C_E \le 5.48910. \tag{25}$$

(ii) On the far range,

$$\sum_{j=0}^{k} \int_3^\infty \frac{u^{k-j}}{(k-j)!} |E_j(u)| \, \mathrm{d}u \le \frac{C_E \Sigma_2}{18} (k+3)^3 3^{-k}. \tag{26}$$

(iii) On the near range the two groups of terms in (16) *are bounded separately:*

$$\sum_{j=0}^{k} \int_0^3 \frac{u^{k-j}}{(k-j)!} |h_j(s(u))| J(u) \, \mathrm{d}u \le C_N 3^{-k} \quad (59 \le k \le 200), \quad C_N \le 367.8436, \tag{27}$$

$$\sum_{j=0}^{k} \int_0^3 \frac{u^{k-j}}{(k-j)!} \big(|h_j(1)| \mathrm{e}^{-u} + |b_j| \mathrm{e}^{-2u} \big) \, \mathrm{d}u \le N_U (k+3)^3 3^{-k}, \qquad N_U \le 26.0962. \tag{28}$$

(iv) $\beta > 0$, and

$$4900 (62)^3 3^{-59} < \frac{\beta}{3} 2^{-59}. \tag{29}$$

(v) Direct interval evaluation gives $g_k > g_{k+1}$ for $1 \le k \le 75$.

Part (v) is not a coefficient bound but a sweep. The range in (27) is two-sided, and its two ends are discharged quite differently: the lower end is harmless because monotonicity below the analytic threshold comes from (v) and never appeals to (31), while the upper end is removed in the proof of Proposition 3.5, where a Cauchy estimate extends the bound past $k = 200$.

The lower restriction is a feature of the method rather than of the inequality. The h_j alternate in sign, so the left-hand side has to be majorized rather than evaluated in closed form, and the majorant is at its loosest where $a(s) = 2|h_1(s)|$ is largest, which is at the left end of the s-range; the resulting bound is correspondingly weakest for small k, where the weights $u^{k-j}/(k-j)!$ put least mass away from that end. It exceeds $C_N 3^{-k}$ exactly for $9 \le k \le 25$, peaking at $699.87 \cdot 3^{-k}$ at $k = 14$, and falls to $34.14 \cdot 3^{-k}$ by $k = 59$. We make no claim about the left-hand side itself outside the stated range: it is the certified bound, not the quantity, that needs the restriction, and $k \ge 26$ would in fact serve. We state $k \ge 59$ because that is the only range in which (31) is ever applied.

The numeral $C_N \le 367.8436$ is carried, rather than the 34.14 actually computed on $k \ge 59$, so that the ledger below is unchanged. It is not idle either: it is the least four-decimal bound on the supremum over all $k \ge 1$ of $3^k P_k^{\mathrm{odd}}(s(3))$, the signed quantity whose coefficient identity earlier drafts of this proof used; that supremum is $367.84350\ldots$, attained at $k = 11$.

Lemma 3.4. *Let $A(z) = \sum_{j \ge 0} a_j z^j$.*
(i) If $\sum_m 3^m |[z^m](1 - z/3)A(z)| \le S$, then $|a_j| \le S 3^{-j}$.
(ii) If $\sum_m 3^m |[z^m](1 - z/3)^2 A(z)| \le S$, then $|a_j| \le S(j+1)3^{-j}$.
(iii) If $\sum_m 3^m |[z^m](1 - z/3)^3 A(z)| \le S$, then

$$\sum_{j=0}^{k} 3^j |a_j| \le \binom{k+3}{3} S. \tag{30}$$

Proof. Write $(1-z/3)^r A(z) = \sum_m w_m z^m$. Then

$$A(z) = \left(\sum_{q\ge 0} \binom{q+r-1}{r-1} 3^{-q} z^q\right)\left(\sum_{m\ge 0} w_m z^m\right).$$

For $r=1$ this gives $|a_j| \le 3^{-j}\sum_{m\le j} 3^m|w_m| \le S3^{-j}$. For $r=2$ it gives $|a_j| \le (j+1)3^{-j}S$. For $r=3$,

$$\begin{aligned}\sum_{j=0}^{k} 3^j|a_j| &\le \sum_{m=0}^{k} 3^m|w_m| \sum_{q=0}^{k-m}\binom{q+2}{2}\\ &= \sum_{m=0}^{k} 3^m|w_m|\binom{k-m+3}{3} \le \binom{k+3}{3}S.\end{aligned}$$

□

Proposition 3.5. *For every $k \ge 59$,*

$$|R_k| \le 4900(k+3)^3 3^{-k}. \tag{31}$$

The threshold is not an artefact of the method but of the near range: the coefficients h_j alternate in sign, so the near-range term has to be estimated through the majorant (34), and the resulting bound exceeds (31) for small k. Nothing is lost, since monotonicity for $k \le 75$ is certified directly and does not use (31); see the proof of Theorem 1.2.

Proof. Write R_k as the sum of the far contribution, the near contribution, and the two coefficient tails of (17)–(18). The far term and the near-range main term are cubic in k; the other two are bounded by constant multiples of 3^{-k}, and since $(k+3)^3 \ge 1$ the cubic factor may be absorbed once, at the end.

Far range $u \ge 3$. The exact error is the sum of the three terms of (16)—no asymptotic series is summed—so the certified Taylor model of (26) applies to the integrand. Its k-dependence is supplied not by that computation, which certifies finitely many numbers, but by Lemma 3.4(iii): applied to the second s-derivative of H_{o} with $S = \Sigma_2$, (30) gives $\sum_{j\le k} 3^j|\partial_s^2 h_j| \le \binom{k+3}{3}\Sigma_2 \le (k+3)^3\Sigma_2/6$. The remaining factor 3 is not a width — the far range $u \ge 3$ is unbounded — but a rate: the far-range error is $O(\mathrm{e}^{-3u})$, which is what the powers of ζ in (24) encode, and $\int_0^\infty u^m \mathrm{e}^{-3u}\,\mathrm{d}u/m! = 3^{-m-1}$. Hence $|R_{k,\mathrm{far}}| \le \frac{C_E\Sigma_2}{18}(k+3)^3 3^{-k}$, the $18 = 6\cdot 3$ recording those two steps.

First tail. By Lemma 3.4(i), $\left|\sum_{j>k} h_j(1)\right| \le \Sigma_0 \sum_{j>k} 3^{-j} = \frac{\Sigma_0}{2}3^{-k}$.

Second tail. Put $B(z) = \partial_s H_{\mathrm{o}}(1,z) + 2\gamma H_{\mathrm{o}}(1,z)$. Multiply by $(1-z/3)^2$ and treat the two summands separately. The first contributes Σ_1 by its definition in Proposition 3.3. For the second, write $(1-z/3)^2 H_{\mathrm{o}} = (1-z/3)\cdot\big[(1-z/3)H_{\mathrm{o}}\big]$; the weighted ℓ^1 norm $\sum_m 3^m|[z^m]\cdot|$ is submultiplicative under this extra factor $(1-z/3)$, whose own norm is 2, so the second summand contributes at most $2\gamma\cdot 2\Sigma_0 = 4\gamma\Sigma_0$ — this is where the 4 comes from, rather than the 2 of the prefactor. Since $\gamma < 0.578$,

$$\sum_{m\ge 0} 3^m|[z^m](1-z/3)^2 B(z)| \le \Sigma_1 + 4\gamma\Sigma_0 < 590,$$

so $|b_j| \le 590(j+1)3^{-j}$ by Lemma 3.4(ii). Summing the resulting geometric series,

$$2^{-k-1}\sum_{j>k}|b_j|2^j \le 590\cdot 2^{-k-1}\sum_{j>k}(j+1)(2/3)^j = 590(k+4)3^{-k},$$

and $(k+4) \le \frac{5}{64}(k+3)^3$ for $k \ge 1$, with equality at $k = 1$.

Near range $0 < u \le 3$. Here $|E_j|$ is at most the true integrand plus the two subtracted main terms, and the latter contribute at most $N_U(k+3)^3 3^{-k}$ by (28). For the former, (12) and the substitution $s = s(u)$, under which $u = \log\zeta(s)$ and $J(u)\,\mathrm{d}u = \mathrm{d}s$, give

$$\sum_{j=0}^{k} \int_0^3 \frac{u^{k-j}}{(k-j)!} |h_j(s(u))| J(u)\,\mathrm{d}u = \int_{s(3)}^{\infty} A_k(s)\,\mathrm{d}s, \qquad A_k(s) = \sum_{j=0}^{k} \frac{(\log\zeta(s))^{k-j}}{(k-j)!} |h_j(s)|. \tag{32}$$

It is A_k, and not the signed sum, that has to be estimated: writing $\Pi(t) = \sum_{p\ge3} p^{-t}$, the z-linear coefficient of $\log H_{\mathrm{o}}$ in (9) is

$$h_1(s) = \Pi(s) - \log\zeta(s) = \log(1-2^{-s}) - \sum_{m\ge2} \frac{\Pi(ms)}{m} < 0, \tag{33}$$

both terms being negative, so the h_j are not of one sign and the identity $\sum_j u^{k-j} h_j(s)/(k-j)! = P_k^{\mathrm{odd}}(s)$ bounds A_k from below rather than above.

We majorize instead. By (33) every coefficient of $\log H_{\mathrm{o}}(s,z) = h_1(s)z + \sum_{m\ge2} \Pi(ms) z^m/m$ beyond the first is positive, so replacing h_1 by $|h_1|$ produces

$$H_{\mathrm{o}}^{+}(s,z) = H_{\mathrm{o}}(s,z) \mathrm{e}^{-2h_1(s)z},$$

a power series with non-negative coefficients dominating $|h_j|$ termwise. Hence, with $a(s) = 2|h_1(s)|$,

$$A_k(s) \le [z^k]\big(\zeta(s)^z H_{\mathrm{o}}^{+}(s,z)\big) = \sum_{i=0}^{k} P_{k-i}^{\mathrm{odd}}(s) \frac{a(s)^i}{i!} =: P_k^{+}(s). \tag{34}$$

To estimate $\int_{s(3)}^{\infty} P_k^{+}$, fix a cut point $s_0 > s(3)$ and treat the two pieces differently. Each is effective for its own reason.

On the compact piece $[s(3), s_0]$, every P_j^{odd} is a Dirichlet series with positive terms and, by (33), a is a positive combination of such series; a sum of products of non-negative decreasing functions is decreasing, so P_k^{+} decreases in s. A finite left-endpoint Riemann sum over $[s(3), s_0]$ is therefore an upper bound for $\int_{s(3)}^{s_0} P_k^{+}$, with no quadrature error to estimate — which is what makes this step certifiable without any analysis of the quadrature rule itself.

On the tail $[s_0, \infty)$ the integral converges at a certified rate. Every odd n with $\Omega(n) = k - i$ satisfies $n \ge 3^{k-i}$, so for $s \ge s_0$, writing $t = s - s_0$, we get $P_{k-i}^{\mathrm{odd}}(s) \le 3^{-(k-i)t} P_{k-i}^{\mathrm{odd}}(s_0)$, while every Dirichlet term of a has $n \ge 2$ and hence $a(s) \le 2^{-t} a(s_0)$ by (33). Since $3^{-x} \le 2^{-x}$ for $x \ge 0$, the ith term of P_k^{+} is multiplied by at most $3^{-(k-i)t} 2^{-it} \le 2^{-kt}$; all terms being non-negative, Tonelli permits termwise integration, and $\int_0^{\infty} 2^{-kt}\,\mathrm{d}t = 1/(k\log 2)$ gives

$$\int_{s_0}^{\infty} P_k^{+}(s)\,\mathrm{d}s \le \frac{P_k^{+}(s_0)}{k\log 2}. \tag{35}$$

Adding the Riemann sum over $[s(3), s_0]$ to (35) bounds $\int_{s(3)}^{\infty} P_k^{+}$, hence $\int_{s(3)}^{\infty} A_k$ by (34). For one k this is a finite computation, and (27) is its outcome for each of the 142 indices $59 \le k \le 200$.

That leaves the dependence on k, which no finite computation can settle, and which has to be reduced exactly as the far range's was. Here the reduction is Cauchy's estimate. Because $H_{\mathrm{o}}^{+}(s,\cdot)$ has non-negative coefficients, its maximum on a circle $|z| = R$ is attained at $z = R$, so for any $R < 3^{s(3)}$ and any $s \ge s(3)$,

$$A_k(s) \le R^{-k} \zeta(s)^R H_{\mathrm{o}}(s,R) \mathrm{e}^{a(s)R} = R^{-k} \prod_{p\ge3} \big(1 - Rp^{-s}\big)^{-1} \mathrm{e}^{a(s)R}, \tag{36}$$

a bound whose only k-dependence is the factor R^{-k}. It cannot be integrated over the whole of $[s(3),\infty)$, since its right-hand side tends to 1 rather than to 0 as $s\to\infty$; so we use it on $[s(3),s_1]$ for a cut point s_1 and fall back on the decay already established beyond it. On $[s(3),s_1]$, (36) gives $\int_{s(3)}^{s_1}A_k\le R^{-k}\Theta_1$ with $\Theta_1=\int_{s(3)}^{s_1}\prod_{p\ge3}(1-Rp^{-s})^{-1}\mathrm{e}^{a(s)R}\,\mathrm{d}s$, finite and independent of k. Beyond s_1, (35) applies with $s_0=s_1$, and bounding $P_k^+(s_1)$ by (36) in turn gives $\int_{s_1}^{\infty}A_k\le R^{-k}\Theta_2/(k\log2)$ with $\Theta_2=\prod_{p\ge3}(1-Rp^{-s_1})^{-1}\mathrm{e}^{a(s_1)R}$, again independent of k. Hence

$$3^k\int_{s(3)}^{\infty}A_k\le\Big(\frac{3}{R}\Big)^k\Big(\Theta_1+\frac{\Theta_2}{k\log2}\Big),$$

Take $R=3.15$, which is below $3^{s(3)}=3.1738\ldots$, and $s_1=8$. The certified values are then $\Theta_1+\Theta_2/(200\log2)\le175346$, while $(3/3.15)^{201}=5.5075\cdot10^{-5}$, so the right-hand side is at most 9.66 for every $k>200$. Therefore

$$\int_{s(3)}^{\infty}A_k(s)\,\mathrm{d}s\le C_N3^{-k}\qquad\text{for all }k\ge59,\tag{37}$$

which is the range in which (31) is used. Neither branch is anywhere near binding: over $59\le k\le200$ the largest value is $34.14\cdot3^{-k}$, at $k=59$, and the Cauchy branch gives at most $9.66\cdot3^{-k}$ beyond, against the $C_N=367.8436$ carried in the ledger.

The companion bound (28) reduces the same way. Here $|h_j(1)|\le\Sigma_03^{-j}$ by Lemma 3.4(i), while $b_j=h_j'(1)+2\gamma h_j(1)$ needs both parts: (ii) gives $|h_j'(1)|\le(j+1)\Sigma_13^{-j}$ and (i) again gives the $2\gamma h_j(1)$ term, so $|b_j|\le\big((j+1)\Sigma_1+2\gamma\Sigma_0\big)3^{-j}$. These turn each integral into a regularized incomplete gamma value, and $P(m,x)\le x^m/m!$ makes the resulting sums telescope: with $U=3$ the first is at most $\Sigma_0\mathrm{e}^93^{-k}$ and the second at most $\frac12\big((k+1)\Sigma_1+2\gamma\Sigma_0\big)\mathrm{e}^93^{-k}$. Both are linear in k where the budget $(k+3)^33^{-k}$ is cubic, so the ratio of this telescoped majorant to the budget decreases in k; since at $k=400$ that ratio is already 12.08, below N_U, the maximum over the finite head $1\le k\le400$ governs. It is attained at $k=7$, far outside the range in which (31) is used.

Substituting the certified constants, the four contributions are

$$|R_k|\le\Big(\underbrace{\frac{5.48910\cdot14414}{18}}_{4395.55}+\underbrace{\frac{44.5987}{2}+\frac{5\cdot590}{64}+C_N+N_U}_{462.34}\Big)(k+3)^33^{-k}<4900(k+3)^33^{-k},$$

with $C_N=367.8436$ and $N_U=26.0962$. □

Proof of Theorem 1.2. The limit follows at once from (21) and Proposition 3.5. For monotonicity,

$$g_k-g_{k+1}=\beta2^{-k-1}+R_k-R_{k+1}.\tag{38}$$

It is enough to prove $|R_m|<\beta2^{-m}/3$ for $m=k,k+1$, because then

$$|R_k|+|R_{k+1}|<\frac{2\beta}{3}2^{-k-1}+\frac{\beta}{3}2^{-k-1}=\beta2^{-k-1}.$$

Define

$$q(k)=\frac{3\cdot4900}{\beta}(k+3)^3(2/3)^k.$$

The certified inequality (29) says $q(59)<1$. Moreover

$$\frac{q(k+1)}{q(k)}=\frac23\left(\frac{k+4}{k+3}\right)^3<1\qquad(k\ge4).$$

Thus $q(k) < 1$ for every $k \geq 59$, and Proposition 3.5 gives $|R_k| < \beta 2^{-k}/3$ throughout that range. Equation (38) proves monotonicity for $k \geq 59$. The certified direct computation covers $1 \leq k \leq 75$, so the two ranges cover every $k \geq 1$. They overlap on the seventeen indices $59 \leq k \leq 75$. □

4. The unrestricted sums: an exact criterion

Write

$$f_k = 1 - 2^{-k}\big(a_1 k^2 + E(k)\big), \qquad a_1 = \frac{d \log 2}{4}. \tag{39}$$

Then

$$f_{k+1} - f_k = 2^{-k-1}\left(a_1(k^2 - 2k - 1) + 2E(k) - E(k+1)\right). \tag{40}$$

Thus $f_k < f_{k+1}$ if and only if

$$a_1(k^2 - 2k - 1) + 2E(k) - E(k+1) > 0. \tag{41}$$

We now derive the decomposition of $E(k)$ used to bound the second term.

Set

$$F(s,z) = \prod_p (1 - zp^{-s})^{-1}, \qquad G_2(s,z) = (s-1)^z (1 - z2^{-s}) F(s,z),$$

and write $c_i = [z^i]G_2(1,z)$. Since $G_2(1,z) = H_{\mathrm{o}}(1,z)$, the power series has radius 3 and

$$\sum_{i \geq 0} c_i = \frac{1}{2}, \qquad \sum_{i \geq 0} 2^i c_i = d. \tag{42}$$

The coefficients are those of Section 3:

$$c_i = h_i(1). \tag{43}$$

Define the absolutely convergent signed moments

$$U = \sum_{i \geq 0} i 2^i c_i, \qquad V = \sum_{i \geq 0} i^2 2^i c_i. \tag{44}$$

Following [3, §3], put

$$I_k = \int_1^2 [z^k] F(s,z)\, \mathrm{d}s, \tag{45}$$

$$I_k' = \sum_{i+j+\ell = k} c_i \int_1^2 \frac{(-\log(s-1))^\ell}{\ell!} 2^{-js}\, \mathrm{d}s. \tag{46}$$

For $m \geq 0$ define

$$A_m = 2^m \sum_{j+\ell = m} \int_1^2 \frac{(-\log(s-1))^\ell}{\ell!} 2^{-js}\, \mathrm{d}s \tag{47}$$

and

$$\delta_m = A_m - 2^{m+1} + \frac{\log 2}{4} m^2, \qquad \Delta(k) = \sum_{i=0}^{k} 2^i c_i \delta_{k-i}. \tag{48}$$

Proposition 4.1. *Let*

$$E_{S_1}(k) = -2^k(I_k - I'_k), \tag{49}$$

$$L_k = 2\sum_{i>k} c_i, \qquad Q_k = \sum_{i>k} 2^i c_i (k-i)^2, \tag{50}$$

$$T_k = f_k - I_k, \tag{51}$$

$$E_{\exp}(k) = 2^k L_k - \frac{\log 2}{4} Q_k - 2^k T_k. \tag{52}$$

Then the identity

$$E(k) = E_{S_1}(k) + \frac{\log 2}{4}(V - 2kU) - \Delta(k) + E_{\exp}(k) \tag{53}$$

holds for every $k \geq 1$.

Proof. From (46) and (47),

$$I'_k = 2^{-k} \sum_{i=0}^{k} 2^i c_i A_{k-i}.$$

Insert the definition of δ_{k-i}:

$$I'_k = 2\sum_{i=0}^{k} c_i - \frac{\log 2}{4} 2^{-k} \sum_{i=0}^{k} 2^i c_i (k-i)^2 + 2^{-k}\Delta(k). \tag{54}$$

By (42) and (44),

$$2\sum_{i=0}^{k} c_i = 1 - L_k,$$

$$\begin{aligned} \sum_{i=0}^{k} 2^i c_i (k-i)^2 &= \sum_{i\geq 0} 2^i c_i (k-i)^2 - Q_k \\ &= dk^2 - 2kU + V - Q_k. \end{aligned}$$

Since $a_1 = \frac{\log 2}{4} d$, substitution into (54) gives

$$I'_k = 1 - 2^{-k}\left(a_1 k^2 + \frac{\log 2}{4}(V - 2kU) - \Delta(k) + 2^k L_k - \frac{\log 2}{4} Q_k \right).$$

Finally $f_k = I'_k + (I_k - I'_k) + T_k$. Comparing with (39) yields (53). □

5. Bounds for the four source terms

5.1. The S_1 term.

Set

$$\alpha_i = \sup_{1\leq t\leq 2} |[z^i]\partial_s G_2(t,z)|, \qquad \alpha_+ = \frac{1}{2}\sum_{i\geq 0} \alpha_i 2^i.$$

The series converges, and a pole location at one value of t is not enough to see it: what is needed is a bound on the coefficients uniform in t.

First, G_2 is analytic in s on a neighbourhood of $[1,2]$, the apparent branch point of $(s-1)^z$ at $s=1$ notwithstanding. Indeed $F(s,z) = \zeta(s)^z H(s,z)$ by (8), so

$$G_2(s,z) = \big((s-1)\zeta(s)\big)^z (1 - z2^{-s}) H(s,z),$$

and $(s-1)\zeta(s)$ is analytic near $s=1$ with value 1 there, the pole of ζ being simple with residue 1; hence $\big((s-1)\zeta(s)\big)^z = \exp\big(z\log((s-1)\zeta(s))\big)$ is analytic there too, and H is analytic by Proposition 2.2. (At $s=1$ this also gives $G_2(1,z) = H_{\mathrm{o}}(1,z)$, as used above.)

The first z-pole of $\partial_s G_2(t,\cdot)$ is at $z = 3^t$, so for $t \in [1,2]$ all of these functions are analytic in $|z| < 3$. Fix r with $2 < r < 3$. Then $M = \sup_{1\le t\le 2}\sup_{|z|=r}|\partial_s G_2(t,z)|$ is finite, the set $[1,2]\times\{|z|=r\}$ being compact and $\partial_s G_2$ continuous on it by the previous paragraph, and Cauchy's estimate on the fixed circle $|z|=r$ gives $\alpha_i \le Mr^{-i}$ uniformly in t. Hence $\sum_{i\ge 0}\alpha_i 2^i \le M(1-2/r)^{-1} < \infty$.

Lemma 5.1. *For every $k \ge 1$,*

$$|E_{S_1}(k)| \le \alpha_+(k+1), \tag{55}$$

and hence

$$|2E_{S_1}(k) - E_{S_1}(k+1)| \le \alpha_+(3k+4). \tag{56}$$

Proof. Since $[z^i]G_2(\cdot,z)$ is complex-valued, the mean value theorem is not available; we integrate instead. By (10) and the analyticity of G_2 in s established above,

$$[z^i]G_2(s,z) - c_i = \int_1^s [z^i]\partial_t G_2(t,z)\,\mathrm{d}t, \qquad \text{so} \qquad |[z^i]G_2(s,z) - c_i| \le (s-1)\alpha_i$$

for $1 \le s \le 2$, by the definition of α_i as a supremum over that interval. Subtracting (46) from the coefficient expansion for I_k and then setting $s = 1 + \mathrm{e}^{-t}$ gives

$$\begin{aligned} |I_k - I'_k| &\le \sum_{i+j+\ell=k} \alpha_i \int_0^\infty \frac{t^\ell}{\ell!} 2^{-j(1+\mathrm{e}^{-t})}\mathrm{e}^{-2t}\,\mathrm{d}t \\ &\le \sum_{i+j+\ell=k} \alpha_i 2^{-j} \int_0^\infty \frac{t^\ell}{\ell!}\mathrm{e}^{-2t}\,\mathrm{d}t \\ &= \sum_{i+j+\ell=k} \alpha_i 2^{-j-\ell-1}. \end{aligned}$$

Multiplying by 2^k and using $k = i+j+\ell$,

$$|E_{S_1}(k)| \le \frac{1}{2}\sum_{i=0}^{k} \alpha_i 2^i (k-i+1) \le \alpha_+(k+1).$$

The triangle inequality now gives

$$|2E_{S_1}(k) - E_{S_1}(k+1)| \le 2\alpha_+(k+1) + \alpha_+(k+2) = \alpha_+(3k+4).$$

□

5.2. **The signed moment term.** The combination of signed moments that has to be bounded is not chosen; it is what the difference of two consecutive source terms produces:

$$2(V - 2kU) - \big(V - 2(k+1)U\big) = V + (2-2k)U. \tag{57}$$

Since this is an identity, bounding the right-hand side is an exact optimization over the rectangle of admissible (U,V), not a termwise absolute-value estimate. That is the reason for evaluating the moments before absolute values are taken: replacing them by $\sum_i 2^i|c_i|$ would discard the cancellation in (57) and cost a factor in k.

Lemma 5.2. *If $|U| \le U_{\max}$ and $|V| \le V_{\max}$, then for $k \ge 2$,*

$$\frac{\log 2}{4}\big(V + (2-2k)U\big) \ge -\frac{\log 2}{4}\big(V_{\max} + (2k-2)U_{\max}\big). \tag{58}$$

Proof. The coefficient of V is 1, while the coefficient $2-2k$ of U is negative. The minimum over the rectangle $[-U_{\max}, U_{\max}] \times [-V_{\max}, V_{\max}]$ is therefore attained at $U = U_{\max}$ and $V = -V_{\max}$, giving (58). □

5.3. **The δ term.** The next lemma supplies the self-contained bound used for $\Delta(k)$.

Lemma 5.3. *For $j \geq 0$ and $u \geq 0$ put*

$$B_j(u) = 2^{-j\mathrm{e}^{-u}} - 1 + j \log 2\, \mathrm{e}^{-u}$$

and

$$\eta_i(j) = \frac{1}{i!} \int_0^\infty u^i \mathrm{e}^{-u} B_j(u)\, \mathrm{d}u.$$

Then $B_j(u) \geq 0$, $\eta_i(j) \geq 0$, and

$$\delta_m = -1 - \frac{\log 2}{4} m + \sum_{i=0}^{m} 2^i \eta_i(m-i). \tag{59}$$

Proof. With $s = 1 + \mathrm{e}^{-u}$ in (47),

$$A_m = \sum_{i=0}^{m} \frac{2^i}{i!} \int_0^\infty u^i \mathrm{e}^{-u} 2^{-(m-i)\mathrm{e}^{-u}}\, \mathrm{d}u.$$

Use $2^{-j\mathrm{e}^{-u}} = 1 - j \log 2\, \mathrm{e}^{-u} + B_j(u)$. The constant term contributes

$$\sum_{i=0}^{m} 2^i = 2^{m+1} - 1.$$

The linear term contributes

$$\begin{aligned} -\log 2 \sum_{i=0}^{m} \frac{2^i (m-i)}{i!} \int_0^\infty u^i \mathrm{e}^{-2u}\, \mathrm{d}u &= -\frac{\log 2}{2} \sum_{i=0}^{m} (m-i) \\ &= -\frac{\log 2}{4} m(m+1). \end{aligned}$$

The remaining term is $\sum_{i=0}^{m} 2^i \eta_i(m-i)$. Subtracting 2^{m+1} and adding $(\log 2) m^2/4$ gives (59). Finally, for $x \geq 0$, $\mathrm{e}^{-x} \geq 1 - x$; applying this with $x = j \log 2\, \mathrm{e}^{-u}$ proves $B_j(u) \geq 0$. □

Lemma 5.4. *For every $m \geq 0$,*

$$|\delta_m| \leq D(m), \qquad D(m) = (\log 2) m \log(m+1) + 1. \tag{60}$$

Proof. For the lower bound, (59) gives $\delta_m \geq -1 - \frac{\log 2}{4} m \geq -D(m)$ for $m \geq 1$, because $\log(m+1) \geq \log 2 > 1/4$; and $\delta_0 = -1 = -D(0)$.

For the upper bound put $\Lambda_m = \sum_{i=0}^{m} 2^i \eta_i(m-i)$. Since $\mathrm{e}^{-x} - 1 + x \leq \min(x, x^2/2)$ for $x \geq 0$, taking $x = (m-i) \log 2\, \mathrm{e}^{-u}$ gives

$$\Lambda_m \leq \int_0^\infty \mathrm{e}^{-u} \sum_{i=0}^{m} \frac{(2u)^i}{i!} \min\left((m-i) \log 2\, \mathrm{e}^{-u}, \frac{(m-i)^2 (\log 2)^2 \mathrm{e}^{-2u}}{2} \right) \mathrm{d}u. \tag{61}$$

Let $v_m = \max\left(0, \log \frac{m+1}{3}\right)$. On $0 \leq u \leq v_m$ use the linear bound together with $\sum_{i=0}^{m} (m-i)(2u)^i/i! \leq m\mathrm{e}^{2u}$, and on $u \geq v_m$ the quadratic bound together with $\sum_{i=0}^{m} (m-i)^2 (2u)^i/i! \leq m^2 \mathrm{e}^{2u}$. Then

$$\Lambda_m \leq m \log 2\, v_m + \frac{m^2 (\log 2)^2}{2} \mathrm{e}^{-v_m}. \tag{62}$$

For $m \geq 2$ we have $v_m = \log \frac{m+1}{3}$ and $\mathrm{e}^{-v_m} = \frac{3}{m+1}$, so (62) and $\frac{3m^2}{2(m+1)} \leq \frac{3}{2}m$ give

$$\begin{aligned}\Lambda_m &\leq m \log 2 \log \frac{m+1}{3} + \frac{3m^2(\log 2)^2}{2(m+1)} \\ &\leq m \log 2 \left(\log(m+1) - \log 3 + \tfrac{3}{2}\log 2\right) \leq m \log 2 \log(m+1),\end{aligned}$$

the last step because $3 > 2^{3/2}$. For $m \leq 1$ we have $v_m = 0$, and (62) reads $\Lambda_m \leq m^2(\log 2)^2/2$, which is at most $m \log 2 \log(m+1)$ since $\frac{m \log 2}{2} \leq \log(m+1)$ there. In both cases

$$\delta_m = -1 - \frac{\log 2}{4} m + \Lambda_m \leq \Lambda_m \leq (\log 2) m \log(m+1) < D(m),$$

which with the lower bound proves (60). □

Lemma 5.5. *If $S^* = \sum_{i \geq 0} 2^i |c_i|$, then, with D as in Lemma 5.4,*

$$|2\Delta(k) - \Delta(k+1)| \leq S^* \big(2D(k) + D(k+1)\big). \tag{63}$$

Proof. Since D is increasing,

$$|\Delta(k)| \leq \sum_{i=0}^{k} 2^i |c_i| D(k-i) \leq S^* D(k).$$

Apply the triangle inequality to $2\Delta(k) - \Delta(k+1)$. □

5.4. **The exponentially small term.** The three terms in (52) are respectively a coefficient tail at $z = 1$, a weighted coefficient tail at $z = 2$, and the integral over $s \geq 2$. The following is the exact numerical input used for them.

Proposition 5.6 (Certified unrestricted input)**.** *The interval computations give*

$$\begin{gathered} d \geq 0.3786949, \qquad S^* \leq 6.9764, \qquad \alpha_+ \leq 15.425661, \\ |U| \leq U_{\max} = 0.08436, \qquad |V| \leq V_{\max} = 1.907462. \end{gathered} \tag{64}$$

For every $k \geq 900$,

$$2^k |T_k| \leq \frac{1.45}{k}, \tag{65}$$

and a contour bound for the regularized generating function $\widetilde{G}_2$ of the c_i,

$$\sup_{|z|=4.2} \left|\widetilde{G}_2(z)\right| \leq 101.245. \tag{66}$$

Together these give

$$|E_{\exp}(k)| \leq \frac{1.45}{k} + 10^{-3} \qquad (k \geq 900), \tag{67}$$

as derived in §5.4. Of the three terms of $E_{\exp}$, two have their dependence on k reduced analytically there, to geometric bounds whose constant is (66)*; only T_k needs a certificate that is itself uniform in k, namely* (65)*. Both are recorded here rather than inside that derivation so that every certified input for the unrestricted case appears in one place. Finally, $f_k < f_{k+1}$ is certified on*

$$6 \leq k \leq 199, \qquad 78 \leq k \leq 899, \qquad 900 \leq k \leq 2439, \tag{68}$$

whose union is the single range $6 \leq k \leq 2439$. These three ranges are not established the same way, and the distinction matters to anyone checking the certificates. On the first, f_k itself is computed as an enclosure for each k and the two enclosures are compared; the quadrature behind those values, and the six error sources entering each, are described in Appendix A. (That

computation in fact covers $2 \le k \le 500$*; only* $6 \le k \le 199$ *is needed here.) On the other two, it is the left-hand side of* (41) *that is evaluated in ball arithmetic at each* k *and found positive — the same quantity whose closed-form lower bound* φ *drives Section 6, but evaluated per* k *rather than bounded uniformly, which is why it closes well below the threshold* 2440 *that the uniform bound requires. Within the middle range two assemblies of that quantity are involved: the refined one for* $78 \le k \le 734$*, and the earlier, coarser one it improves on for* $735 \le k \le 899$*.*

The coefficient tails in (67) are obtained only after the pole at $z = 3$ has been removed. Put

$$\widetilde{G}_2(z) = (1 - z/3)G_2(1, z) = \sum_{m \ge 0} w_m z^m.$$

The certified contour bound is (66). Hence $|w_m| \le 101.245\, 4.2^{-m}$, and from $G_2(1, z) = (1 - z/3)^{-1}\widetilde{G}_2(z)$ we obtain

$$|c_i| \le 101.245\, 3^{-i} \sum_{m=0}^{i} (3/4.2)^m < 354.36\, 3^{-i}. \tag{69}$$

By (43) the c_i are the coefficients $h_i(1)$, so Lemma 3.4(i) sharpens (69) to $|c_i| \le \Sigma_0 3^{-i}$, dispensing with the contour bound; the weaker form suffices below. With $C = 354.36$,

$$\begin{aligned} 2^k|L_k| &\le C(2/3)^k, \\ \frac{\log 2}{4}|Q_k| &\le \frac{\log 2}{4} C(2/3)^k \sum_{r \ge 1} r^2 (2/3)^r = \frac{30C \log 2}{4}(2/3)^k. \end{aligned}$$

Both terms are less than 10^{-3} in total for $k \ge 900$, and both bounds are geometric in k, so no computation is needed to make them uniform. The third term is the one that needs a certificate, namely (65); substituting it into (52) proves (67), and no asymptotic O-term enters the final criterion. The first two terms are in fact below 10^{-3} from $k \ge 37$ on, so the threshold 900 comes from T_k alone.

6. Assembly of the unrestricted theorem

For $k \ge 900$ define

$$\begin{aligned} R(k) = {} & \frac{\log 2}{4}\big((2k-2)U_{\max} + 2V_{\max} + 4U_{\max}\big) \\ & + S^*\big(2D(k) + D(k+1)\big) + \alpha_+(3k+4) + 3\left(\frac{1.45}{k} + 10^{-3}\right), \end{aligned} \tag{70}$$

and

$$\varphi(k) = a_1(k^2 - 2k - 1) - R(k). \tag{71}$$

Proposition 6.1. *For* $k \ge 900$*,*

$$2E(k) - E(k+1) \ge -R(k). \tag{72}$$

Consequently $\varphi(k) > 0$ *implies* $f_k < f_{k+1}$*.*

Proof. Apply (53) at k and $k+1$. The S_1 contribution is bounded by (56). The signed moment contribution is

$$\frac{\log 2}{4}\big(V + (2-2k)U\big),$$

which is bounded below by (58). The first line of (70) is slightly larger than the resulting quantity, since

$$V_{\max} + (2k-2)U_{\max} \le (2k-2)U_{\max} + 2V_{\max} + 4U_{\max}.$$

The Δ contribution is bounded by (63). Finally, (67) gives

$$|2E_{\exp}(k) - E_{\exp}(k+1)| \le 2\left(\frac{1.45}{k} + 10^{-3}\right) + \left(\frac{1.45}{k+1} + 10^{-3}\right) \le 3\left(\frac{1.45}{k} + 10^{-3}\right).$$

Adding the four lower bounds proves (72). Equation (40) then gives

$$f_{k+1} - f_k \ge 2^{-k-1}\varphi(k),$$

which proves the last assertion. □

Lemma 6.2. *With the certified constants in Proposition 5.6, $\varphi(2440) > 1155$.*

Proof. Substitution in (71) gives the following certified one-sided bounds:

$$\begin{aligned} a_1(2440^2 - 2\cdot 2440 - 1) &> 390371.6,\\ \frac{\log 2}{4}\big(4878U_{\max} + 2V_{\max} + 4U_{\max}\big) &< 72.1,\\ S^*\big(2D(2440) + D(2441)\big) &< 276166.8,\\ \alpha_+(3\cdot 2440 + 4) &< 112977.6,\\ 3\left(\frac{1.45}{2440} + 10^{-3}\right) &< 0.005. \end{aligned}$$

Therefore

$$\varphi(2440) > 390371.6 - 72.1 - 276166.8 - 112977.6 - 0.005 > 1155.$$

The same substitution at $k = 2433$ gives $\varphi(2433) > 135$, so the criterion is already in force there. Direct evaluation puts the sign change just below that, at $\varphi(2432) < 0 < \varphi(2433)$; we do not rely on it, since certifying $\varphi(2432) < 0$ would need a *lower* bound on $R(2432)$, whereas every constant in (64) is one-sided the other way. Nothing below requires the first index: we carry the threshold at 2440, which is one past the end of (68), so that the sweep and the criterion meet without a gap. □

Lemma 6.3. *The function $k \mapsto \varphi(k)$ is strictly increasing for integers $k \ge 2440$.*

Proof. First,

$$a_1\big((k+1)^2 - 2(k+1) - 1 - (k^2 - 2k - 1)\big) = a_1(2k-1).$$

Moreover

$$\begin{aligned} D(m+1) - D(m) &= \log 2\left(\log(m+2) + m\log\left(1 + \frac{1}{m+1}\right)\right)\\ &\le \log 2\big(\log(m+2) + 1\big). \end{aligned}$$

The last term in (70) decreases with k. Hence

$$\begin{aligned} R(k+1) - R(k) \le{}& \frac{\log 2}{2}U_{\max} + 3\alpha_+\\ &+ S^*\log 2\big(2\log(k+2) + \log(k+3) + 3\big). \end{aligned} \tag{73}$$

Define, for real $x \ge 2440$,

$$\begin{aligned} M(x) = a_1(2x-1) &- \frac{\log 2}{2}U_{\max} - 3\alpha_+\\ &- S^*\log 2\big(2\log(x+2) + \log(x+3) + 3\big). \end{aligned}$$

Then

$$M'(x) = 2a_1 - S^*\log 2\left(\frac{2}{x+2} + \frac{1}{x+3}\right) > 0,$$

because the certified lower bound for d gives $2a_1 > 0.1312$, whereas the subtracted term is < 0.006 for $x \geq 2440$. Direct substitution gives $M(2440) > 146$. Thus $M(k) > 0$ for every $k \geq 2440$. Together with (73), this yields $\varphi(k+1) - \varphi(k) > 0$. □

Proof of Theorem 1.1. By Lemmas 6.2 and 6.3, $\varphi(k) > 0$ for every $k \geq 2440$. Proposition 6.1 therefore gives $f_k < f_{k+1}$ on that range. The certified ranges in (68) cover every integer from 6 through 2439. Together the analytic and certified ranges cover all $k \geq 6$. □

7. A general principle

Conjecture 7.1. Let $\mathcal{P}$ be a set of primes with least element q and thin complement,

$$\sum_{p \notin \mathcal{P}} \frac{\log p}{p} < \infty, \tag{74}$$

and let $f_k^{\mathcal{P}}$ be the sum of $1/(n \log n)$ over integers n with $\Omega(n) = k$ and all prime factors in $\mathcal{P}$. The sequence $f_k^{\mathcal{P}}$ is eventually increasing if and only if $q = 2$.

Some hypothesis is needed. For $\mathcal{P} = \{2\}$ one has $f_k^{\mathcal{P}} = 1/(2^k k \log 2)$, and for $\mathcal{P} = \{2, 3\}$ the sequence again decreases to 0; in both cases $q = 2$ and the conclusion fails. What goes wrong is that a sparse $\mathcal{P}$ leaves $\sum_{p \in \mathcal{P}} p^{-s}$ bounded as $s \to 1^+$, so the generating function has no $\zeta(s)^z$ factor and $f_k^{\mathcal{P}}$ decays geometrically whatever q is.

Those examples, however, are already excluded by the weaker condition $\sum_{p \notin \mathcal{P}} 1/p < \infty$, and that condition is what governs the size of the limit. The candidate limit is $\prod_{p \notin \mathcal{P}}(1 - 1/p)$, which is the value $H_{\mathcal{P}}(1, 1)$ of (76) below, and it is a convergent product with positive value precisely when $\sum_{p \notin \mathcal{P}} 1/p < \infty$, and 0 otherwise. That it is indeed the limit is proved below for $q \geq 3$, in Proposition 7.2(iv); for $\mathcal{P}$ all primes it is Lichtman's $f_k \to 1$ [2]; and for the two sparse examples the product is 0, as are their limits. So the hypothesis separates a sequence with a positive limit, whose monotonicity is decided by a correction term, from one that decays to 0 — the sparse examples being the degenerate case of one product rather than a separate phenomenon.

It is worth saying separately why (74) carries the weight $\log p$ instead. It is not needed for convergence: the product below converges locally uniformly for *any* set of primes, since $\sum_p p^{-\sigma} < \infty$ for $\sigma > 1/2$. What needs the weight is the s-derivative at $s = 1$, where the constant multiplying 2^{-k} lives. That constant involves $\sum_{p \notin \mathcal{P}} \frac{\log p}{p-1}$, which is finite precisely under (74) and can diverge under the weaker condition: taking the complement to be blocks of primes in $[X_j, X_j^{1+\eta_j}]$ with $\eta_j \to 0$ slowly and $\log X_j$ growing geometrically makes $\sum_{p \notin \mathcal{P}} 1/p$ converge while $\sum_{p \notin \mathcal{P}} (\log p)/p$ diverges. We have not verified that such a $\mathcal{P}$ violates the conclusion, so we state the hypothesis in the safe direction: strengthening it makes the conjecture claim less.

The singularity structure is the one the two theorems above exploit. On $|z| < q^\sigma$ the product $\prod_{p \in \mathcal{P}}(1 - zp^{-s})^{-1}$ converges directly, by the estimate in the proof of Proposition 2.2, so its first z-singularity is at $z = q^s$; the factorization

$$\prod_{p \in \mathcal{P}} (1 - zp^{-s})^{-1} = \prod_{p} (1 - zp^{-s})^{-1} \prod_{p \notin \mathcal{P}} (1 - zp^{-s})$$

then relates it to $\zeta(s)^z H_{\mathcal{P}}(s, z)$ on the smaller disc $|z| < 2^\sigma$ where the middle product also converges. At $s = 1$ the point $z = 2$ lies on the singular circle exactly when $q = 2$ — and $z = 2$ is distinguished here not because 2 is prime but because it is where the γe^{-2u} term of J in Lemma 2.3 evaluates the regularized product. If $q \geq 3$ the extraction at $z = 2$ is interior and gives a constant multiple of 2^{-k}, as in Section 3; if $q = 2$ it is at the boundary, and the

mechanism that then produces a factor k^2 is the one identified by Gorodetsky, Lichtman and Wong [3] for $\mathcal{P} = \{\text{all primes}\}$.

In the interior case $q \geq 3$ the regularized product does supply the control the extraction needs, and the constant can be identified. Write

$$L_p(s,z) = (1-p^{-s})^z(1-zp^{-s})^{-1}, \qquad \log L_p(s,z) = \sum_{m\geq 2} \frac{z^m - z}{m} p^{-ms}, \tag{75}$$

the second identity being the computation in the proof of Proposition 2.2. Define $H_{\mathcal{P}}$ by the regularized product, as (8) and (9) define H and H_{o}:

$$H_{\mathcal{P}}(s,z) = \prod_{p\notin\mathcal{P}} (1-p^{-s})^z \prod_{p\in\mathcal{P}} (1-zp^{-s})^{-1}(1-p^{-s})^z = \prod_{p\notin\mathcal{P}} (1-p^{-s})^z \prod_{p\in\mathcal{P}} L_p(s,z), \tag{76}$$

which is (8) when $\mathcal{P}$ is all primes and (9) when $\mathcal{P} = \{p : p \geq 3\}$. For $s > 1$, where $\zeta(s)^z$ is finite and non-zero, (7) extends to $\mathcal{P}$ as

$$\sum_{k\geq 0} P_k^{\mathcal{P}}(s) z^k = \prod_{p\in\mathcal{P}} (1-zp^{-s})^{-1} = \zeta(s)^z H_{\mathcal{P}}(s,z), \qquad P_k^{\mathcal{P}}(s) = \sum_{\substack{\Omega(n)=k \\ p|n \Rightarrow p\in\mathcal{P}}} n^{-s}, \tag{77}$$

by the same computation, since $\zeta(s)^{-z} = \prod_p (1-p^{-s})^z$ there. It is (76) and not the factor $\zeta(s)^{-z}\prod_{\mathcal{P}}(1-zp^{-s})^{-1}$ of (77) that is used below: the latter is 0 at $s=1$, where every constant in this section is evaluated, while the former is finite and non-zero there. Put $h_j^{\mathcal{P}}(s) = [z^j]H_{\mathcal{P}}(s,z)$.

Proposition 7.2. *Let $\mathcal{P}$ be a set of primes with least element $q \geq 3$ and complement $\mathcal{Q}$ satisfying* (74)*, and fix $R \in (2,3)$, which is possible with $R < q$ since $q \geq 3$. Then:*

(i) *for every real $s \geq 1$ both products in* (76) *converge absolutely and locally uniformly on $|z| < q^s$, where $H_{\mathcal{P}}(s,\cdot)$ is therefore analytic and zero-free;*

(ii) *$M := \sup_{s\geq 1}\sup_{|z|=R}|H_{\mathcal{P}}(s,z)| < \infty$, and hence $|h_j^{\mathcal{P}}(s)| \leq MR^{-j}$ for all $j \geq 0$ and all $s \geq 1$;*

(iii) *each $h_j^{\mathcal{P}}$ is C^1 on $[1,\infty)$ with $\sup_{s\geq 1}|(h_j^{\mathcal{P}})'(s)| \leq M'R^{-j}$ for some $M' < \infty$, and*

$$\tfrac{1}{2}\partial_s \log H_{\mathcal{P}}(1,2) = -\sum_{p\in\mathcal{P}} \frac{\log p}{(p-1)(p-2)} + \sum_{p\in\mathcal{Q}} \frac{\log p}{p-1}, \tag{78}$$

both sums finite, the second precisely by (74)*;*

(iv) *consequently*

$$f_k^{\mathcal{P}} = \prod_{p\in\mathcal{Q}} \Big(1-\frac{1}{p}\Big) + c_{\mathcal{P}} 2^{-k} + o(2^{-k}), \qquad c_{\mathcal{P}} = H_{\mathcal{P}}(1,2)\big(\gamma + \tfrac{1}{2}\partial_s \log H_{\mathcal{P}}(1,2)\big), \tag{79}$$

with $H_{\mathcal{P}}(1,2) = \prod_{p\in\mathcal{P}}(1-p^{-1})^2(1-2p^{-1})^{-1}\prod_{p\in\mathcal{Q}}(1-p^{-1})^2 \in (0,\infty)$.

Proof. (i) Fix $s \geq 1$ and a compact $K \subset \{|z| < q^s\}$, say $|z| \leq R' < q^s$. For the product over $\mathcal{P}$, (75) gives $\log L_p(s,z) = O_{R'}(p^{-2s})$ uniformly on K by the estimate in the proof of Proposition 2.2, whose hypothesis $|zp^{-s}| < 1$ holds because $|z| \leq R' < q^s \leq p^s$ for every $p \in \mathcal{P}$; and $\sum_p p^{-2s} < \infty$ since $s \geq 1$. For the product over $\mathcal{Q}$, $z\log(1-p^{-s}) = O_{R'}(p^{-s})$ and

$$\sum_{p\in\mathcal{Q}} p^{-s} \leq \sum_{p\in\mathcal{Q}} \frac{1}{p} \leq \frac{1}{2} + \frac{1}{\log 3}\sum_{p\in\mathcal{Q}} \frac{\log p}{p} < \infty \qquad (s \geq 1),$$

the comparison $1/p \leq (\log p)/(p\log 3)$ holding for every $p \geq 3$ and the term $p = 2$ — which lies in $\mathcal{Q}$ for every $\mathcal{P}$ with $q \geq 3$, and for which the comparison fails since $\log 2 < 1$ — being split off and bounded by $1/2$. So (74) gives $\sum_{p\in\mathcal{Q}} 1/p < \infty$, which is all that is used here.

Both products therefore converge absolutely and locally uniformly, so each is analytic in z on $|z| < q^s$; and neither vanishes, because the convergence of the logarithms exhibits each as the exponential of a finite quantity.

(ii) The two estimates in (i), applied with $R' = R < 3 \le q \le q^s$, depend on s only through $p^{-s} \le p^{-1}$, so

$$\sup_{s\ge1}\sup_{|z|=R}|\log H_{\mathcal{P}}(s,z)| \le C_R\sum_p p^{-2} + R\sum_{p\in\mathcal{Q}}\log\frac{1}{1-1/p} < \infty,$$

which bounds M; the bound on $h_j^{\mathcal{P}}$ is Cauchy's estimate on $|z| = R$.

(iii) Differentiating (76) logarithmically term by term,

$$\partial_s \log H_{\mathcal{P}}(s,z) = -\sum_{p\in\mathcal{P}}\log p\sum_{m\ge2}(z^m - z)p^{-ms} + z\sum_{p\in\mathcal{Q}}\frac{\log p}{p^s-1}. \tag{80}$$

For $s \ge 1$ and $|z| \le R$ the first double sum is dominated termwise by $C_R(\log p)p^{-2}$ and the second by $R(\log p)/(p-1)$, summable over $\mathcal{Q}$ exactly by (74). Both bounds are independent of s, so by the Weierstrass M-test the differentiated series converges uniformly there and the differentiation is legitimate; its sum is continuous on $[1,\infty)\times\{|z|\le R\}$, being a uniformly convergent series of continuous functions, and $\partial_s H_{\mathcal{P}} = H_{\mathcal{P}}\partial_s \log H_{\mathcal{P}}$ is bounded there. Since $[z^j]$ is a Cauchy integral over the circle $|z| = R$, which may be held fixed while s varies, differentiation under the integral sign gives $(h_j^{\mathcal{P}})'(s) = [z^j]\partial_s H_{\mathcal{P}}(s,z)$ exactly as in (10), and Cauchy's estimate then bounds it by $M'R^{-j}$. At $z = 2$ the inner sum in (80) is

$$\sum_{m\ge2}(2^m-2)p^{-m} = \frac{4}{p(p-2)} - \frac{2}{p(p-1)} = \frac{2}{(p-1)(p-2)},$$

which gives (78) after halving.

(iv) Write $P_k^{\mathcal{P}}(s) = \sum n^{-s}$ over the n counted by $f_k^{\mathcal{P}}$. The proof of Lemma 2.1 used nothing about the index set beyond non-negativity of the summands, so it applies verbatim and gives $f_k^{\mathcal{P}} = \int_1^\infty P_k^{\mathcal{P}}(s)\,\mathrm{d}s$. Expanding $\zeta(s)^z = \mathrm{e}^{uz}$ with $u = \log\zeta(s)$ and taking the Cauchy product with $H_{\mathcal{P}}(s,z) = \sum_j h_j^{\mathcal{P}}(s)z^j$ gives $P_k^{\mathcal{P}}(s) = \sum_{j=0}^k h_j^{\mathcal{P}}(s)u^{k-j}/(k-j)!$, which is (12) with $\mathcal{P}$ in place of the odd primes; changing variables to u as in (13), by Lemma 2.3, then gives

$$f_k^{\mathcal{P}} = \int_0^\infty \sum_{j=0}^k h_j^{\mathcal{P}}(s(u))\frac{u^{k-j}}{(k-j)!}J(u)\,\mathrm{d}u.$$

Split the integral at $u = 1$ and write $s_1 = s(1)$, so that $s_1 > 1$. The range $u \in (0,1]$ is $s \ge s_1$, where the contribution is $\int_{s_1}^\infty P_k^{\mathcal{P}}(s)\,\mathrm{d}s$ and may be bounded without any of the above: every n counted by $P_k^{\mathcal{P}}$ has $n \ge q^k \ge 3^k$, so $P_k^{\mathcal{P}}(s) \le 3^{-k(s-s_1)}P_k^{\mathcal{P}}(s_1)$ for $s \ge s_1$, whence

$$\int_{s_1}^\infty P_k^{\mathcal{P}}(s)\,\mathrm{d}s \le \frac{P_k^{\mathcal{P}}(s_1)}{k\log 3} \le \frac{2^{-k}}{k\log 3}\prod_{p\in\mathcal{P}}(1-2p^{-s_1})^{-1} = O\Big(\frac{2^{-k}}{k}\Big) = o(2^{-k}),$$

the middle step because the generating function (7) for $\mathcal{P}$ has non-negative coefficients, so $P_k^{\mathcal{P}}(s_1) \le 2^{-k}\sum_j P_j^{\mathcal{P}}(s_1)2^j$, and the product converges because $2 < 3^{s_1} \le q^{s_1}$. On $[1,\infty)$ the expansions (11) apply with remainders that are $O(\mathrm{e}^{-3u})$ *uniformly* there: each remainder times e^{3u} is continuous on $[1,\infty)$ and bounded as $u \to \infty$ by Lemma 2.3, hence bounded on $[1,\infty)$. With $\int_0^\infty \frac{u^m}{m!}\mathrm{e}^{-au}\,\mathrm{d}u = a^{-(m+1)}$,

$$\int_1^\infty \frac{u^m}{m!}J\,\mathrm{d}u = 1 + \gamma 2^{-m} + O(3^{-m}), \qquad \int_1^\infty \frac{u^m}{m!}\varepsilon J\,\mathrm{d}u = \tfrac{1}{2}2^{-m} + O(3^{-m}), \tag{81}$$

the restriction to $[1,\infty)$ costing at most $\int_0^1 u^m/m!\,\mathrm{d}u \le 1/(m+1)!$ in each, which is smaller than the error terms already present and is absorbed into them throughout. Write

$$h_j^{\mathcal{P}}(s(u)) = h_j^{\mathcal{P}}(1) + \varepsilon(u)\,(h_j^{\mathcal{P}})'(1) + r_j(u), \qquad |r_j(u)| \le \varepsilon(u)\,\Omega(\varepsilon(u))\,R^{-j},$$

where

$$\Omega(\delta) = \sup_{|z|=R}\ \sup_{1\le s\le 1+\delta} |\partial_s H_{\mathcal{P}}(s,z) - \partial_s H_{\mathcal{P}}(1,z)|.$$

Here $r_j(u) = \int_1^{s(u)} \big((h_j^{\mathcal{P}})'(t) - (h_j^{\mathcal{P}})'(1)\big)\,\mathrm{d}t$, and Cauchy's estimate on $|z| = R$ applied to $\partial_s H_{\mathcal{P}}(t,\cdot) - \partial_s H_{\mathcal{P}}(1,\cdot)$ bounds the integrand by $\Omega(t-1)R^{-j}$, which gives the stated bound on r_j since Ω is non-decreasing and $s(u) - 1 = \varepsilon(u)$. Moreover $\Omega(\delta) \to 0$ as $\delta \to 0$: by (iii) the function $\partial_s H_{\mathcal{P}}$ is continuous on $[1,\infty) \times \{|z| \le R\}$, hence uniformly continuous on the compact set $[1,2] \times \{|z| = R\}$, which covers $\delta \le 1$. The three groups of terms then contribute as follows. By (ii) with $R > 2$, the tails $\sum_{j>k}$ of $\sum_j h_j^{\mathcal{P}}(1)$ and of $2^{-k}\sum_j h_j^{\mathcal{P}}(1)2^j$ are $O(R^{-k})$ and $O(2^{-k}(2/R)^k)$, both $o(2^{-k})$, and the $O(3^{-m})$ terms contribute $O\big(3^{-k}\sum_{j\le k} MR^{-j}3^j\big) = O(R^{-k}) = o(2^{-k})$, the middle equality using $R < 3$, so that the geometric sum is dominated by its last term; so the first group gives $H_{\mathcal{P}}(1,1) + \gamma H_{\mathcal{P}}(1,2)2^{-k} + o(2^{-k})$. The second group is handled the same way, using (iii) in place of (ii): its tail $\sum_{j>k}$ of $2^{-k}\sum_j (h_j^{\mathcal{P}})'(1)2^j$ is $O(2^{-k}(2/R)^k)$ and its $O(3^{-m})$ terms are $O(R^{-k})$, so it gives $\frac{1}{2}\partial_s H_{\mathcal{P}}(1,2)2^{-k} + o(2^{-k})$; the series $\sum_j (h_j^{\mathcal{P}})'(1)2^j$ converges to $\partial_s H_{\mathcal{P}}(1,2)$ because $R > 2$. For the third, let $\delta > 0$ and choose U with $\Omega(\varepsilon(U)) < \delta$, which is possible since $\varepsilon(U) \to 0$. On $[U,\infty)$ the terms are at most δR^{-j} times the second integral of (81), contributing $O(\delta 2^{-k})$; on $[1,U]$ the integrand is $O_U(1)$ times $u^{k-j}/(k-j)!$, and $\int_1^U u^m/m!\,\mathrm{d}u \le U^{m+1}/(m+1)!$, so that range contributes $O_U\big(\sum_{j\le k} R^{-j}U^{k-j+1}/(k-j+1)!\big) = O_U(R^{-k}) = o(2^{-k})$. As δ was arbitrary the third group is $o(2^{-k})$. Collecting the two surviving coefficients gives $c_{\mathcal{P}} = \gamma H_{\mathcal{P}}(1,2) + \frac{1}{2}\partial_s H_{\mathcal{P}}(1,2)$, which is (79). Finally $H_{\mathcal{P}}(s,1) = \prod_{p\in\mathcal{Q}}(1-p^{-s})$ by (76), since $L_p(s,1) = 1$, giving the limit; and the stated product for $H_{\mathcal{P}}(1,2)$ is (76) at $(s,z) = (1,2)$. Its factors are positive because $p \ge q \ge 3$ on $\mathcal{P}$, and the product is non-zero — which positivity of the factors alone would not give — by the convergence in (i) at $(s,z) = (1,2)$, legitimate since $2 < 3 \le q$. □

Positivity of $c_{\mathcal{P}}$ — which by (79) is what makes the sequence decrease — is now a corollary rather than an open point.

Corollary 7.3. *$c_{\mathcal{P}} > 0$ for every $\mathcal{P}$ with $q \ge 3$ satisfying (74); indeed $c_{\mathcal{P}} \ge H_{\mathcal{P}}(1,2)\,\beta/d$, and $f_k^{\mathcal{P}}$ is eventually strictly decreasing.*

Proof. Put $a_p = \frac{\log p}{(p-1)(p-2)} \ge 0$ and $b_p = \frac{\log p}{p-1} \ge 0$ for $p \ge 3$. Since $q \ge 3$ we have $2 \in \mathcal{Q}$ and $\mathcal{P} = \{p \ge 3\} \setminus (\mathcal{Q} \setminus \{2\})$, with $b_2 = \log 2$, so (78) rearranges exactly — for every admissible $\mathcal{Q}$, finite or infinite — as

$$\tfrac{1}{2}\partial_s \log H_{\mathcal{P}}(1,2) = \log 2 - \sum_{p\ge 3} a_p + \sum_{p\in\mathcal{Q}\setminus\{2\}} (a_p + b_p),$$

all three sums converging by (74). The last sum has non-negative terms, so the right side is least when it is empty, that is at $\mathcal{Q} = \{2\}$ and $\mathcal{P} = \{p : p \ge 3\}$, where $\gamma + \frac{1}{2}\partial_s \log H_{\mathcal{P}}(1,2)$ equals $\gamma + \log 2 - \sum_{p\ge 3} \frac{\log p}{(p-1)(p-2)} = \beta/d = 0.410970\ldots$, positive by (5). Since $H_{\mathcal{P}}(1,2) > 0$ by Proposition 7.2(iv), $c_{\mathcal{P}} > 0$, and (79) gives $f_k^{\mathcal{P}} - f_{k+1}^{\mathcal{P}} = c_{\mathcal{P}} 2^{-k-1} + o(2^{-k}) > 0$ for all large k. □

So the single inequality $\beta > 0$ of (5) settles the whole interior case. In particular Corollary 7.3 is one of the two implications of Conjecture 7.1: it shows that $f_k^{\mathcal{P}}$ is eventually strictly decreasing

whenever $q \geq 3$, so an admissible $\mathcal{P}$ with $f_k^{\mathcal{P}}$ eventually increasing must have $q = 2$. What remains is the converse, and the boundary case $q = 2$ lies outside Proposition 7.2, whose parts (ii), (iii) and (iv) all require $R > 2$ and so an admissible $R < q$; part (i) needs neither, and (76) itself is a definition for any $\mathcal{P}$ whatever.

What can be said is that the boundary case reduces to the single case of all primes, and that the object the reduction turns on is an interior one. For $q = 2$ we have $2 \in \mathcal{P}$, and separating that one factor in (76) gives the exact identity

$$H_{\mathcal{P}}(s,z) = \frac{H_{\mathcal{P}\setminus\{2\}}(s,z)}{1 - z2^{-s}}, \tag{82}$$

since $L_2 = (1-2^{-s})^z(1-z2^{-s})^{-1}$ and moving 2 into the complement contributes exactly the compensating factor $(1-2^{-s})^z$. For $\mathcal{P}$ all primes this is $H = H_{\mathrm{o}}/(1-z2^{-s})$: the whole z-singularity of the unrestricted regularized product at $z = 2^s$ is that one explicit pole, and its numerator is the odd product (9). In general the numerator $H_{\mathcal{P}\setminus\{2\}}$ belongs to a set of primes with least element at least 3, so Proposition 7.2 applies to *it*: it is analytic and zero-free on $|z| < 3^s$, in particular at the critical point $(s,z) = (1,2)$, where by (83) below it takes the value $d\prod_{p\in\mathcal{Q}}(1-2/p) > 0$.

Comparing (76) for $\mathcal{P}$ with the same for all primes gives

$$\frac{H_{\mathcal{P}}(s,z)}{H(s,z)} = \frac{H_{\mathcal{P}\setminus\{2\}}(s,z)}{H_{\mathrm{o}}(s,z)} = \prod_{p\in\mathcal{Q}}(1-zp^{-s}), \tag{83}$$

the first equality on $|z| < 2^s$, where all four functions are defined, and the second on the larger disc $|z| < 3^s$, both sides there being analytic by Proposition 7.2(i) applied to $\mathcal{P}\setminus\{2\}$ and to the odd primes. It is the second equality that is used at $z = 2$, which is interior to $|z| < 3^s$ for every $s \geq 1$. Both H and $H_{\mathcal{P}}$ blow up at $(1,2)$, which lies on the boundary of the smaller disc; the point of (83) is that their *ratio* does not, being equal there to a ratio of two interior objects. Since $q = 2$ forces $2 \notin \mathcal{Q}$, every $p \in \mathcal{Q}$ is at least 3, so the product is one of factors in $(0,1)$ which under (74) converges to the non-zero value $\prod_{p\in\mathcal{Q}}(1-2/p)$.

Now the k^2 coefficient is known in the all-primes case, and it is exactly the numerator's value. By (2) and (3) it is $a_1 = d(\log 2)/4$, and $d = H_{\mathrm{o}}(1,2)$ — (3) is (9) at $(s,z) = (1,2)$ — so

$$a_1 = \tfrac{1}{4}(\log 2)\, H_{\mathrm{o}}(1,2), \tag{84}$$

the value of the pole's numerator at the critical point times a constant carrying no arithmetic of its own. The contrast with the interior case is sharp and is the reason to think (84) is structural rather than a coincidence: the interior constant $\beta = d(\gamma + \frac{1}{2}\partial_s \log H_{\mathrm{o}}(1,2))$ of (5) does involve the s-derivative of the same numerator, and a prime sum with it, whereas a_1 involves neither. Granting that (84) holds with $H_{\mathcal{P}\setminus\{2\}}$ in place of H_{o} — that is, that the residue at the pole sees the numerator through its value — the k^2 coefficient for a general admissible $\mathcal{P}$ with $q = 2$ is

$$a_1^{\mathcal{P}} = \tfrac{1}{4}(\log 2)\, H_{\mathcal{P}\setminus\{2\}}(1,2) = a_1 \prod_{p\in\mathcal{Q}}\Big(1 - \frac{2}{p}\Big) > 0, \tag{85}$$

and no cancellation occurs for any of them.

This is testable, and the prediction (85) is the ratio $\prod_{p\in\mathcal{Q}}(1-2/p)$. For $\mathcal{Q} = \{3\}$ the quantity

$$\Big(f_k^{\mathcal{P}} - \prod_{p\in\mathcal{Q}}(1-1/p)\Big)\Big/(f_k - 1),$$

computed from (77) and Lemma 2.1 independently of everything above, rises monotonically over $24 \leq k \leq 44$ to 0.31209 against a predicted $1/3$; the approach is from below with a correction

of order $1/k$, so the value at $k = 44$ is not the thing to read. Fitting $L + a/k + b/k^2$ through $k = 36, 40, 44$ gives $L = 0.333227$, and through the two other windows $k = 24, 32, 40$ and $k = 28, 36, 44$ gives 0.333171 and 0.333447 — stable, and within $5 \cdot 10^{-4}$ of 1/3. Two caveats keep this from being quoted more sharply. A three-parameter fit through three points has no residual, so the agreement is not an error estimate: fitting $L + a \log k/k + b/k$ instead gives 0.329517 and the one-term fit $L + a/k$ gives 0.336577, so the spread across reasonable bases, about 1%, is the honest uncertainty. And the choice of the $1/k$ basis is not independent of the conclusion, since (2) carries a subleading $O(k \log(k+1))$ whose logarithm cancels in the ratio only if its coefficient also scales by $\prod(1 - 2/p)$ — a stronger form of what is being tested. The stability of the $1/k$ fits across windows is evidence for that basis, not a proof of it. The centring constant $\prod_{p \in \mathcal{Q}}(1 - 1/p)$ is likewise not supplied by Proposition 7.2, which excludes $q = 2$; it is the value (76) gives at $z = 1$, and at $\mathcal{Q} = \varnothing$ it is 1, which is Lichtman's $f_k \to 1$.

The constants of Proposition 7.2(iv) are checked separately, and the two checks should not be conflated. The constant *formulas* specialize correctly to the certified values: at $\mathcal{Q} = \{2\}$ they return $H_{\mathcal{P}}(1, 2) = 0.378695020$ against $d = 0.3786950315$ and $c_{\mathcal{P}} = 0.155632714$ against β, both to about 10^{-7}, the precision of the prime sums involved. The *sequence* $f_k^{\mathcal{P}}$ converges to its asymptotic much more slowly, and how slowly depends on q: at $\mathcal{Q} = \{2\}$, where the subleading term is the $O(k^3 3^{-k})$ one of (4) with the constant 4900, the ratio of $(f_k^{\mathcal{P}} - c_0)2^k$ to $c_{\mathcal{P}}$ is still 1.13 at $k = 14$, the largest index computed; at $\mathcal{Q} = \{2, 3\}$, where $q = 5$, it is 1.003. So it is the $q \geq 5$ runs that test (79) end to end, and $\mathcal{Q} = \{2\}$ that calibrates the formulas against a case the paper proves. Both computations are in the repository, as `genP_check.py` and `boundary_check.py`. Unlike everything else in this paper that is computed, they use ordinary floating-point arithmetic and not ball arithmetic: they are checks on the formulas of this section, carrying no certificate, and nothing elsewhere in the paper or in the formalization depends on them.

Three things are therefore unverified in the boundary case, and it is worth separating them. First, that an expansion $f_k^{\mathcal{P}} = c_0 + O(k^2 2^{-k})$ with a genuine k^2 leading term exists at all for a general admissible $\mathcal{P}$ with $q = 2$; Proposition 7.2 does not reach that case, and (2) is the single case $\mathcal{Q} = \varnothing$. Second, that its coefficient is (85). Third, failing that, something weaker still suffices: if the coefficient is $A\Phi(1, 2) + B\,\partial_s\Phi(1, 2)$ for the numerator $\Phi_{\mathcal{P}} = H_{\mathcal{P}\setminus\{2\}}$ with A, B independent of $\mathcal{P}$, then because $\partial_s \log \Phi_{\mathcal{P}} = \partial_s \log H_{\mathrm{o}} + \partial_s \log E_{\mathcal{Q}}$ with $E_{\mathcal{Q}}(s, z) = \prod_{p \in \mathcal{Q}}(1 - zp^{-s})$, the coefficient for $\mathcal{P}$ is $E_{\mathcal{Q}}(1, 2)$ times

$$A\Phi(1, 2) + B\,\partial_s\Phi(1, 2) + B\,\Phi(1, 2) \sum_{p \in \mathcal{Q}} \frac{2 \log p}{p - 2},$$

the sum being $\partial_s \log E_{\mathcal{Q}}(1, 2)$, whose terms are positive because $p \geq 3$ on $\mathcal{Q}$ and which converges precisely by (74) — the third distinct task that hypothesis performs here, after making the limit in (79) positive and the 2^{-k} coefficient finite. Since $E_{\mathcal{Q}}(1, 2) > 0$, such a coefficient vanishes only if that sum takes one exact value κ fixed by A, B and the all-primes data, so non-cancellation would fail at most on a single level set of an explicit functional of $\mathcal{Q}$, and nowhere at all if $\kappa \leq 0$. Against (84), which has $B = 0$, that case is expected to be empty.

The present paper proves the two cases $\mathcal{P} = \{\text{all primes}\}$ and $\mathcal{P} = \{p : p \geq 3\}$, which are extremal among admissible sets in the sense just described.

8. Certification and formalization

What the formalization establishes is an *implication*. The certified quantities enter as fields of six hypothesis structures — `NumericalFacts`, `NumericalFactsFull`, `OddNumericalFacts`, `OddCoeffFacts`, `OddTailFacts`, `OddDomFacts`, with 13, 21, 4, 8, 5 and 5 fields respectively —

and no instance of any of them is constructed. Lean therefore checks that those hypotheses imply the theorems, and nothing in the kernel checks that the hypotheses are true, or even that they are mutually consistent: an unsatisfiable field would make the theorems assuming it vacuous rather than false, and would not be detected. That the imported intervals agree with what Arb computed is verified outside Lean, by the script `verify.py` against the certificate. The reader should take the Lean development as discharging the reductions below and nothing else; the trusted base is the structure fields, and it is listed in Table 1.

With that said, the division is as follows. Formalized in Lean 4 without `sorry` are Lemmas 2.1 and 2.3, the coefficient identities (13) and (53), the assembly of the ranges in both theorems, and the lemmas named in the rest of this section.

Two items a reader would expect to find among them are not there, and we state the position precisely. First, Proposition 2.2: a radius-specific consequence of it is proved, on the ball $|z| < 3$ for $s \geq 1$, but its operative content (10) — that $\mathrm{d}/\mathrm{d}s$ commutes with $[z^j]$ — is not, and the s-differentiability of the coefficient functions is itself a hypothesis, as field `g_deriv` of `NumericalFactsFull` in the full case and as `deriv_exists`, `deriv2_exists` of `OddCoeffFacts` in the odd case. No interval computation can establish a differentiability claim, and none is attempted. Second, Proposition 3.5: its statement is imported as field `remainder` of `OddNumericalFacts` rather than derived from Proposition 3.3; the derivation is the proof given in Section 3 above, which has not been formalized. The Lean development contains the intended route to it, but that route still rests on assumed coefficient and dominating-sequence data of its own, so it does not discharge the hypothesis.

Likewise the source estimates of Section 5 are assumed, not proved, by the theorem named below: `NumericalFacts` carries $|2E(k) - E(k+1)| \leq R(k)$ as the single field `source_bound`, which is the aggregate of that whole section. Only the second formal version discussed below derives it, from the four source terms separately.

It should also be said that the trusted base is not a list of intervals. Some fields are intervals; others are quantified or analytic statements that no interval computation could discharge: a `Summable` hypothesis (`OddCoeffFacts.fam_summable`), two one-sided limits as $s \downarrow 1$ (`tendsto_one` and `tendsto_deriv`), several suprema quantified over all m, k or θ (`alpha_dom`, `Sstar_dom`, `UV_bound`, `M0_bound`, `Eexp_bound`), a certification predicate over all k (`OddDomFacts.dom_cert`), and the sweeps, which are finitely checkable but are $\forall$-statements rather than single enclosures. And six of the constants above — Σ_0, Σ_1, Σ_2, C_E, C_N, N_U — do not appear in the Lean development at all, in any form: the odd-remainder route that would consume them takes them as abstract parameters and is not what either odd theorem is proved from. For those six the agreement between manuscript and formalization that this section asserts has no content, and the reader should rely on the Arb certificates instead.

The numerical propositions are Proposition 3.3 and Proposition 5.6. Their statements are finite systems of interval inequalities, established by Arb/FLINT through its Python bindings. The Lean formalization imports the resulting intervals as hypotheses and proves that they imply Theorem 1.1 and both assertions of Theorem 1.2; the three formal statements are `erdos_sums_increase_from_six`, `fOdd_decreasing` and `fOdd_tendsto_half`. The differentiability field belongs to `NumericalFactsFull` and so is a hypothesis of the second, primitive version of the full theorem rather than of `erdos_sums_increase_from_six`, which does not mention it. The limit $g_k \to \frac{1}{2}$ was formerly left unformalized, the monotonicity argument making no use of it; it is now proved there as well, and from the same hypothesis, since (4) already bounds $|g_k - \frac{1}{2} - \beta 2^{-k}|$ by a multiple of $(k+3)^3 3^{-k}$, and both that and 2^{-k} vanish.

For Theorem 1.1 the hypotheses are not only the interval bounds of Proposition 5.6. Proposition 6.1 enters as a hypothesis in one of the two formal versions, and is proved in the other, from

the primitive certified quantities rather than from the aggregate bound (72). (The bound (67) is not what the primitive route replaces; it remains a hypothesis, `NumericalFactsFull.Eexp_bound`, in both versions.) What the formalization proves from these is the passage to $f_k < f_{k+1}$ for $k \geq 2440$, together with the assembly of the four ranges.

Lemmas 6.2 and 6.3 were formerly hypotheses of that development as well. They are now proved in it, as `phiBound_pos_at_2440` and `phiBound_mono_of_bounds`, from upper bounds on the parameters alone: φ is an explicit expression in k, so neither lemma is a statement about primes, contours or limits, and what the formalization had been assuming was arithmetic. Proving them needed one quantity the certified hypotheses did not carry, the bound $|U| \leq U_{\max}$ of (64), which is now among them. The sharpness required is not uniform: the base case turns on $\log 2441$ to a tenth of a percent, where $\log 2441 \leq 12 \log 2$ is already too weak, while the monotonicity step needs only $\log 2441 \leq 12 \log 2$ itself.

For the odd theorem the lower bound on γ was likewise a hypothesis and is now proved, as `gamma_lower_bound`, from $\mathrm{harmonic}(n) - \log(n+1) < \gamma$ at $n = 8191$, chosen so that $n+1 = 2^{13}$ and the logarithm is $13 \log 2$ exactly.

The constants d of (3) and ρ were hypotheses of both developments and are now proved in them, as `dConst_lower_bound`, `dConst_lower_bound_sharp` and `rhoConst_upper_bound`. What makes this possible is that the Euler factors are rational and exceed 1,

$$\Big(1 - \tfrac{2}{p}\Big)^{-1}\Big(1 - \tfrac{1}{p}\Big)^{2} = \frac{(p-1)^2}{p(p-2)} = 1 + \frac{1}{p(p-2)},$$
$$\Big(1 - \tfrac{3}{p}\Big)^{-1}\Big(1 - \tfrac{1}{p}\Big)^{3} = \frac{(p-1)^3}{p^2(p-3)} = 1 + \frac{3p-1}{p^2(p-3)},$$

the second because $(p-1)^3 - p^2(p-3) = 3p-1$. The constant regularized by the second display is

$$\rho = -\frac{1}{9}\prod_{p \geq 5}\Big(1 - \frac{3}{p}\Big)^{-1}\Big(1 - \frac{1}{p}\Big)^{3}, \tag{86}$$

which is recorded here only so that every constant proved in the formalization has a counterpart in the text; it is used nowhere in the proofs above. A product of factors each at least 1 is bounded below by the product over any finite subset of them, with no tail estimate entering, so a one-sided bound on such a constant is a finite computation in the kernel even where the two-sided enclosure is not. Only one-sided bounds are used: d enters through $a_1 = d \log 2/4$ and through $\beta = d(\log 2 - S + \gamma)$, in both cases from below, and ρ is not used at all. The bounds proved are weaker than those of Proposition 5.6, which are unchanged. The 94 primes $p \leq 499$ give $d \geq 0.3785913$, which suffices for $a_1 \geq 0.0656$; the 668 primes $p \leq 4999$ give $d \geq 0.3786823$, which is what $\beta \geq 0.1556$ requires; and the 93 primes $5 \leq p \leq 499$ give $\rho \leq -0.1747885$. In each case the requirement is computed against the bound for $\log 2$ available in the formalization rather than against $\log 2$ itself, the two differing by enough to change how many primes are needed.

The same device does not reach S. There the bound required is an upper one on a sum of positive terms, so the tail cannot be discarded, and the summands carry $\log p$ rather than being rational. The budget is narrow: with d at the bound just proved, $\beta \geq 0.1556$ permits $S \leq 0.8593986$, while the partial sum over $p < 2 \cdot 10^6$ already reaches 0.8593917, so the admissible tail is below $7 \cdot 10^{-6}$ and primes beyond about $1.5 \cdot 10^5$ are needed. Accordingly S remains a certified hypothesis, as do the sweeps and the Cauchy estimates of Propositions 3.3 and 5.6.

Two constants are stated above more conservatively than the computation requires, so that the text and the formalization carry the same values: the threshold 2440 of Lemma 6.2 in

place of 2433, and (69) in place of the sharper bound that Lemma 3.4(i) supplies. Neither changes the ranges covered. Two further discrepancies should be recorded rather than left for the reader to find. The certified odd sweep is stated above for $1 \le k \le 75$, whereas the field `OddNumericalFacts.sweep_low` assumes only $1 \le k \le 58$; since the analytic branch begins at $k = 59$ that is all the assembly needs, and assuming less is the safe direction, but the two ranges are not the same. And the field `NumericalFacts.Gamma_bound`, carrying $\Gamma \le 104.52$, has no counterpart above and no consumer in the development either; it is retained only so that the certified package and the text can be compared field by field.

TABLE 1. What the Lean development proves, what it assumes, and what it does not mention. The assumed rows are the trusted base: no instance of any structure is constructed, so these are not checked in the kernel. The last row is not part of that base at all — those six constants enter the Lean development only through comments, so for them there is no formal counterpart to compare the manuscript against.

Statement	Status	Lean name
Theorem 1.1	proved	`erdos_sums_increase_from_six`
Theorem 1.2, monotonicity	proved	`fOdd_decreasing`
Theorem 1.2, limit	proved	`fOdd_tendsto_half`
$\varphi(2440) > 0$, not the stronger Lemma 6.2	proved	`phiBound_pos_at_2440`
φ non-decreasing on $k \ge 2440$, not strictly, as Lemma 6.3 says	proved	`phiBound_mono_of_bounds`
γ from below	proved	`gamma_lower_bound`
d from below	proved	`dConst_lower_bound`, `_sharp`
ρ from above	proved	`rhoConst_upper_bound`
Proposition 3.5	assumed	`OddNumericalFacts.remainder`
Section 5 aggregate	assumed	`NumericalFacts.source_bound`
(67)	assumed	`NumericalFactsFull.Eexp_bound`
g_i differentiable on $(1, 2)$	assumed	`NumericalFactsFull.g_deriv`
∂_s, ∂_s^2 of coefficients exist	assumed	`OddCoeffFacts.deriv_exists`, `deriv2_exists`
S^* from above	assumed	`NumericalFacts.Sstar_bound`
S of (5), enclosed	assumed	`OddNumericalFacts.Sodd_enclosure`
$\lvert R_k\rvert$'s constant 4900	assumed	`OddNumericalFacts.MR_bound`
Sweeps	assumed	`OddNumericalFacts.sweep_low` and companions
α_+, U, V	assumed	`alpha_bound`, `U_bound`, `V_bound`
$\Gamma \le 104.52$	assumed, unused	`NumericalFacts.Gamma_bound`
Σ_0, Σ_1, Σ_2, C_E, C_N, N_U	*not in Lean*	see below

For the odd theorem, the coefficient bounds in (25) are obtained by cancelling the pole at $z = 3$ before applying coefficient recursion or Cauchy bounds. The range $0 < u \le 3$ is evaluated directly, while the range $u \ge 3$ uses the Taylor model made explicit in Proposition 3.1.

Σ_2 is not a single computed number but a sum of two, and since it carries 4395.55 of the 4857.88 in the ledger of Proposition 3.5 it is worth saying which. The head $j \le 16$ is a direct interval recursion on the coefficients over 1600 boxes in s, giving 8423.91; a Cauchy estimate would be far lossier there, because it is dominated by the maximum over the circle and discards the cancellation among the true coefficients. The tail $j > 16$ is then exactly where a Cauchy estimate is cheap: with $\sup_{|z|=4.5}$ of the pole-free part at most $1.9673 \cdot 10^6$, the tail is at most that times $(3/4.5)^{17}(1 - 3/4.5)^{-1}$, which is 5989.94. The two sum to 14413.85, so the stated $\Sigma_2 \le 14414$ holds with 0.16 to spare — the narrowest margin of any constant here except C_E. Both halves are recorded in the repository, and the 40 boxes in s behind the tail are coarse enough that it is likely several-fold loose; that direction is safe, and the budget closes either way.

The constant C_E is certified on the whole of $(1, s(3)]$, and in fact on its closure: the four quantities entering (24) are ratios in which the pole of ζ at $s = 1$ cancels, so each extends continuously to $s = 1$, and the Taylor model below is centred at $1 + E/2$ with radius $E/2$ and therefore covers the closed interval $[1, s(3)]$ on which Σ_2 takes its own supremum. Interval evaluation on boxes of s loses that cancellation, since each occurrence of $s - 1$ is widened independently, and the enclosures diverge however finely $(1, s(3)]$ is partitioned. Expanding ζ as a power series in $\sigma = s - s_0$ about $s_0 = 1 + E/2$, where $E = s(3) - 1$, makes the cancellation symbolic. The resulting series are evaluated on $|\sigma| \le E/2$, which is the whole range; their coefficients fall off geometrically, and truncating after nine terms contributes less than 10^{-20}.

The constant C_N requires $\Pi(t) = \sum_{p \ge 3} p^{-t}$ at $t = s, 2s, \dots, 200s$ for $s = s(3)$. At $s \approx 1.05$ these converge too slowly to truncate: the Rosser–Schoenfeld bound applied beyond 10^8 still overestimates the tail twofold, and that excess compounds through the coefficient recursion. Instead Π is evaluated by the Möbius series $\Pi(t) = \sum_{n \ge 1} \mu(n) n^{-1} \log \zeta(nt) - 2^{-t}$, which requires no prime enumeration. The recursion itself is run at 600 bits, since cancellation costs about one digit every 2.3 steps over the 200 terms used; at 1200 bits the head agrees to every printed digit.

For the unrestricted theorem, the s-uniform coefficient bounds defining α_+ are obtained by carrying s as an interval over a partition of $[1, 2]$. The signed moments U and V are evaluated before absolute values are taken, for the reason given at (57).

Availability and versions. Since the contribution is in part a certification, the artefacts and the versions they were produced with are part of the claim. The Lean development, the Arb scripts, their logs and the certificate are at `https://github.com/jaideepsaipadhi/erdos-sums-almost-primes`. The formalization is built with Lean 4, toolchain `leanprover/lean4:v4.33.0-rc1`, against Mathlib at revision `79d0395a1825`; the permitted axioms are exactly `propext`, `Classical.choice` and `Quot.sound`, which is checked by `#print axioms` on each named theorem. The numerics use python-flint 0.9.0 against FLINT 3.6.0. The certificate records the parameters of every run together with a SHA-256 hash of the numerical payload, and `verify.py` checks the imported intervals against it.

Appendix A. Certified quadrature

Since $1/(n\log n) = \int_1^\infty n^{-s}\,\mathrm{d}s$, we have $f_k = \int_1^\infty P_k(s)\,\mathrm{d}s$, and the substitution $s = 1 + \mathrm{e}^{-u}$ turns this into

$$f_k = \int_{-\infty}^{\infty} F(u)\,\mathrm{d}u, \qquad F(u) = P_k(1+\mathrm{e}^{-u})\mathrm{e}^{-u}, \tag{87}$$

the integrand being analytic on the strip $|\mathrm{Im}\,u| < \pi/2$. The integral is split into a truncated range $[u_-, u_+]$ with $u_- = -5.10$ and $u_+ = 3.5k_{\max}$, taken once for the largest k of the run and so uniform in k, and the two tails beyond it. The truncated range is divided into panels of width $H = 0.8$, each mapped affinely to $[-1,1]$ and integrated by Gauss–Legendre quadrature with $N = 52$ nodes at 1400 bits of precision; the number of panels therefore grows linearly with $k_{\max}$.

Four of the six error sources in the certificate are arithmetic: ball arithmetic throughout, the explicit truncation balls in the Möbius series for the prime zeta function, the discretization bound derived below, and interval-Newton enclosures for the Gauss nodes and weights. The remaining two are the tails of (87), which are bounded as follows and are not negligible by inspection.

For the left tail, $u \le u_-$ corresponds to $s \ge s_- = 1+\mathrm{e}^{5.10} = 165.02$. There $P_1(s) = \sum_p p^{-s} \le 2.78\cdot 2^{-s}$, since $P_1(s)2^s = 1 + \sum_{p\ge 3}(2/p)^s$ decreases in s and is already below 2.78 at $s = 2$; and $P_k(s) \le P_1(s)^k$, because expanding $P_1(s)^k$ counts every n with $\Omega(n) = k$ at least once. Hence $P_k(s) \le 2.78^k 2^{-ks}$, and

$$\int_{-\infty}^{u_-} |F|\,\mathrm{d}u = \int_{s_-}^{\infty} P_k(s)\,\mathrm{d}s \le \frac{2.78^k 2^{-ks_-}}{k\log 2}. \tag{88}$$

For the right tail, $P_k(s) \le (\log\zeta(s))^k/k!$ for $s > 1$ [2, Lemma 5.6], so with $u = \log\zeta(s)$ the integral is dominated by $\int_{u_+}^\infty u^k\mathrm{e}^{-u}\,\mathrm{d}u/k!$; the resulting incomplete-Gamma integral obeys $\Gamma(k+1,u_+)/k! \le 2u_+^k\mathrm{e}^{-u_+}/k!$ provided $u_+ \ge 2k+2$, so

$$\int_{u_+}^{\infty} |F|\,\mathrm{d}u \le \frac{2u_+^k\mathrm{e}^{-u_+}}{k!}. \tag{89}$$

The hypothesis $u_+ \ge 2k+2$ holds throughout, since $u_+ = 3.5k_{\max}$ and $k \le k_{\max}$. Both bounds are added to every reported enclosure. They are largest at $k = k_{\max}$, and even there they are far below the differences $f_{k+1} - f_k \asymp a_1 k^2 2^{-k}$ that the sweep has to resolve: for $k_{\max} = 199$ the right-hand side of (89) is below 10^{-100} while the smallest difference in range exceeds 10^{-57}.

For f analytic inside the Bernstein ellipse E_ρ and bounded there by M, the N-node Gauss rule satisfies

$$\left| \int_{-1}^{1} f(x)\,\mathrm{d}x - \widehat{I}_N \right| \le \frac{64}{15}\,\frac{M\rho^{-2N}}{1-\rho^{-2}} \tag{90}$$

[5, Thm. 4.5], where the rule is indexed so that I_n uses $n+1$ nodes and the stated exponent $\rho_B^{-(2n+2)}$ is the ρ_B^{-2N} above; the bound appears as [6, Thm. 19.3]. Here ρ_B is the Bernstein-ellipse parameter, unrelated to the arithmetic constant ρ of Section 8. The implementation evaluates the right side of (90) with the interval-scaling factor $H/2$ and a further factor 10 of margin. The ellipse is taken with semi-minor axis $\delta = 1.45 < \pi/2$ in the u variable, giving $\rho_B \approx 7.385$, and M is bounded using $|P_k(s)| \le P_k(\mathrm{Re}\,s)$ together with the monotonicity of P_k on the reals.

The per-panel bound is not monotone in the panel's position, and it would be wrong to locate its maximum by inspection: on the right-hand side $F(u) \approx u^k\mathrm{e}^{-u}/k!$ by (87), which increases

until $u \approx k$ and only then decays, so M rises across the bulk of the range before falling. What is used is therefore the maximum over all panels and all $k \le k_{\max}$, which the certificate records; it is 1.1×10^{-87}, and so is negligible beside the widths actually reported. It is also, at $k_{\max} = 199$, some twenty orders of magnitude larger than either tail (88)–(89), so the discretization bound and not the truncation is what limits the accuracy of the f_k.

The pinned coefficient sweeps use primes up to 10^5 and 300 bits of precision, with 1600 boxes in s for α_+ and 800 boxes for the second-derivative sweep. The certificate records these parameters together with a SHA-256 hash of the numerical payload.

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Statement on the use of AI

The analytic reductions, the interval-arithmetic certificates, the Lean development and the text of this paper were all produced with extensive use of an AI assistant (Anthropic's Claude). I chose the problem, directed the work throughout, and take full responsibility for every claim made here, including those I reached with that assistance.

In more detail, the assistant was used to derive and check the reductions of Sections 2–6; to write and run the Arb/FLINT scripts that produce the certified constants of Propositions 3.3 and 5.6; to write the Lean development described in Section 8; and to draft and revise this manuscript. It was also used adversarially, to look for errors in work already believed finished. That is how the defect in the near-range step of Proposition 3.5 was found: the step bounded a sum of absolute values by a signed coefficient identity, which is invalid because the h_j alternate in sign. It is the reason that proposition is stated on $k \ge 59$.

Two limits on what the foregoing should be taken to mean. The numerical claims do not rest on the assistant's reliability: they are certified by interval arithmetic and are reproducible from the scripts and logs in the repository cited in Section 8. The Lean development, by contrast, establishes an implication from assumed hypotheses rather than a closed proof, as Section 8 and Table 1 set out in detail.

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